ธนาคารแห่งประเทศไทย
BANK OF THAILAND

# MONETARY POLICY REPORT

September

2019



# Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.


The Monetary Policy Committee
September 2019

| | |
|---|---|
| Mr. Veerathai Santiprabhob | Chairman |
| Mr. Mathee Supapongse | Vice Chairman |
| Mr. Paiboon Kittisrikangwan | Member |
| Mr. Sethaput Suthiwart-Narueput | Member |
| Mr. Kanit Sangsubhan | Member |
| Mr. Subhak Siwaraksa | Member |
| Mr. Somchai Jitsuchon | Member |

# Monetary Policy in Thailand

## Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprises the governor and two deputy governors, as well as four distinguished external members representing various sectors of the economy, with the aim of ensuring that monetary policy decisions are effective and transparent.

## Monetary Policy Objective

The MPC sets monetary policy to promote the objective of supporting sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

## Monetary Policy Target

The Cabinet approved the annual average headline inflation target of 2.5 $\pm$ 1.5 percent as the target for the medium term and for 2019. The inflation target is to assure the general public that the MPC will take necessary policy actions to return headline inflation to the target within an appropriate time horizon without jeopardizing growth and macro-financial stability. In the event that headline inflation deviates from the target, the MPC shall explain the reasons behind the target breach to the Minister of Finance and the public, together with measures taken and estimated time to bring inflation back to the target.

## Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the policy interest rate to signal the monetary policy stance.

## Evaluation of Economic Conditions and Forecasts

The Bank of Thailand takes into account information from all sources, the macroeconomic model, data from each economic sector, as well as surveys of large enterprises, together with small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and cover all aspects, both at the macro and micro levels.

## Monetary Policy Communication

Recognizing the importance of monetary policy communication to the public, the MPC employs various channels of communication, both in Thai and English, such as (1) organizing a press statement at 14:00 on the day of the Committee meeting, (2) publishing edited minutes of the MPC meeting two weeks after the meeting, and (3) publishing the *Monetary Policy Report* every quarter.

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# Executive Summary

## Monetary Policy Conduct in the Third Quarter of 2019

The Thai economy was expected to expand at a lower rate than previously assessed and below potential due to a decline in exports which affected domestic demand. Headline inflation in 2019 was projected to be lower than previously assessed and below the lower bound of the inflation target due to lower-than-expected energy prices and core inflation. However, inflation was expected to rise toward the target next year. Meanwhile, there remained risks in the financial system that might pose vulnerabilities to financial stability in the future. **The Committee voted 5 to 2 to cut the policy rate by 0.25 percentage point from 1.75 to 1.50 percent at the meeting on August 7, 2019 and voted unanimously to maintain the policy rate at 1.50 percent at the meeting on September 25, 2019.** At the most recent meeting, the Committee viewed that accommodative monetary policy stance should support the rise of headline inflation toward target, and would contribute to the continuation of economic growth amid heightened risks mainly from the external front. Nevertheless, the Committee remained concerned about baht appreciation against trading partner currencies, which could affect the economy to a larger degree amid heightened external risks. The Committee thus saw the need to closely monitor developments of exchange rates and capital flows, and would consider implementing additional measures at an appropriate timing if necessary. With regard to addressing financial stability risks, given softening economic outlook and prolonged low interest rates in the period ahead, the Committee viewed that microprudential and macroprudential measures should be appropriately combined and targeted.

Looking ahead, the Committee would be data-dependent by closely monitoring the developments of the outlook for growth, inflation, and financial stability, together with associated risks, especially the impacts from trade tensions, in deliberating appropriate monetary policy going forward. The Committee would stand ready to use policy tools as appropriate. Nevertheless, the Thai economy continued to face structural problems, which would affect competitiveness and growth outlook going forward. This should be firmly addressed by all related parties.

## Assessment of the Economic and Financial Outlook as the Basis for Policy Formulation

### 1. Global Economy

**The global economy was expected to expand at a slower pace** due to intensifying trade tensions between the U.S. and China weighing on production, exports, and private investment. However, private consumption continued to be an important driver, underpinned by robust employment, while more accommodative monetary policy stance and additional government stimulus measures would support growth of advanced economies in the period ahead. **The Committee thus revised down growth projection of Thailand's trading partner economies to 2.8 percent in 2019 and 2.7 percent in 2020. Nonetheless, the probability of trading partner economies underperforming the baseline projection remained** due to (1) uncertainties from prolonged trade tensions, which could undermine consumption and investment outlook, (2) uncertainties pertaining to the recovery of electronics cycle and exports, as well as (3) additional geopolitical risks which could increase volatility in the financial and commodity markets.

**Many central banks cut their policy rates, the majority of them shifting toward more dovish stance to shore up the economy.** The Federal Reserve (Fed) cut its policy rate again in September 2019 to 1.75 – 2.00 percent. Another policy rate cut was expected by the end of this year with another cut in 2020. The European Central Bank (ECB) announced a package of additional monetary policy easing measures in September. The Bank of Japan (BOJ) would likely ease monetary policy further should the economy experience a significant slowdown. Other regional central banks, including the Reserve Bank of Australia (RBA), Bank of Korea (BOK), the Reserve Bank of India (RBI), Bangko Sentral ng Pilipinas (BSP) and Bank Indonesia (BI) lowered their policy rates to support weakening economic and inflation outlook.

**Emerging markets (EMs) continued to experience volatile capital flows.** Capital outflows from EMs occurred due to (1) investor concerns about the Fed's monetary policy stance being less dovish than the markets anticipated, (2) intensifying trade tensions, (3) heightening geopolitical risks, as well as (4) more accommodative monetary policy of many EM central banks, including Thailand. However, capital flows returned to EMs at the end of the third quarter after concerns over trade tensions alleviated and advanced economy central banks shifted towards a more dovish monetary policy stance.

## 2. Financial Conditions and Financial Stability

**Thailand's overall financial conditions remained accommodative** as reflected by the real policy rate that remained at a low level relative to those of other EMs. Interest rates in Thailand's financial markets, including short-term government bond yields, corporate bond yields, and some commercial banks' lending rates declined in tandem with the policy rate cut. However, long-term government bond yields abruptly fell due to both internal and external factors. Overall financing activities continued to expand from the previous quarter. However, private credit exhibited slower growth mainly due to business loans extended to manufacturing, trade, and service sectors. Credit extended to households moderated, especially mortgage and auto leasing loans. The baht appreciated against the U.S. dollar from the end of the previous quarter on account of concerns over global economic slowdown as well as intensifying and prolonged trade tensions. In particular, after the U.S. announced additional tariffs on Chinese imports and designated China as a currency manipulator, safe-haven asset prices rose such as gold, and the Japanese yen and Thai baht strengthened. The real effective exchange rate (REER) index also appreciated from the last quarter.

**There remained risks in the financial system that could pose vulnerabilities to financial stability in the future. However, some risks abated due to the implementation of targeted prudential measures and the economic slowdown.** For instance, risks in the real estate sector had been partly addressed through the revised loan-to-value (LTV) measure, as reflected by (1) more tightening lending standards among financial institutions, especially for loans extended to the same purchaser acquiring the second and subsequent residential units, (2) subdued speculation in the property market, (3) lower residential prices encouraging affordability of some real demand, and (4) subsequent adjustments by real estate developers to cope with softening demand. **On the other hand, some risks had not improved**, including (1) weakening ability of households and SMEs to cope with negative income shocks, (2) rising risks of oversupply in the real estate market following domestic and foreign demand slowdown, and (3) the search-for-yield behavior which could lead to underpricing of risks, particularly among saving cooperatives and large corporates.

## 3. Economic and Inflation Outlook

**The Thai economy was expected to expand at a slower pace at 2.8 and 3.3 percent in 2019 and 2020, respectively**, which would be lower than the previous assessment in the *Monetary Policy Report* and below potential. This was due to a slowdown in both external and domestic demand. Merchandise and services exports were expected to exhibit slower growth than previously assessed due to further slowdown in global trade volume and trading partner economies. In addition, merchandise exports contraction began to impact private consumption and private investment. Meanwhile, public expenditure would grow at a slower rate owing partly to delays in state-owned enterprise investment projects. The annual average of headline inflation was projected to be lower, particularly in 2019 due to falling energy prices and slower core inflation in line with demand-pull pressures. Nonetheless, headline inflation was projected to increase toward the inflation target in 2020 in line with the economic outlook and cost-push inflationary pressures from the prospects of minimum wage increase.

**The value of merchandise exports in 2019 was expected to contract**, mainly due to the decline in export volume following intensifying and prolonged trade tensions. Export prices were projected to fall in line with global crude oil prices. In the period ahead, merchandise exports would recover thanks to the relocation of

production base to Thailand in previous periods. Moreover, the U.S. would import more categories of Thai exports to substitute Chinese imports.

**Exports of services were projected to decelerate** consistent with the outlook of most foreign tourist nationalities and lower spending per head amid the global economic slowdown. However, government measures to boost tourism through extending the exemption of visa-on-arrival (VOA) fees from the end of October 2019 to April 2020 would attract more foreign tourists.

**Private consumption was expected to be weighed down** by weakened purchasing power. This was due to greater impacts of exports on employment and working hours in export-related manufacturing sectors. Farm income was projected to decline next year from the high base effect of agricultural prices due to natural disasters this year. In addition, elevated household debt may exert additional pressures on consumption, especially when household income were restrained. Despite all these, consumption would be shored up in the near term by government economic stimulus measures during the second half of 2019.

**Private investment would expand at a lower rate than previously assessed** in line with export contraction and domestic demand slowdown. However, government stimulus measures would bolster private investment to some extent. Looking ahead, private investment would benefit from additional relocation of production base to Thailand as businesses seek to avoid the impact of intensifying and prolonged trade tensions. Moreover, investor confidence was expected to pick up due to continued infrastructure investment in the Eastern Economic Corridor (EEC) as well as the public-private partnership (PPP) projects.

**Public spending would support economic growth to a lesser extent than previously assessed** due to delays in state-owned enterprise investment projects. In addition, there was a change in budget structure of the Annual Budget Expenditure Act B.E. 2563 (A.D. 2020) that led to a smaller proportion of capital expenditure and a larger proportion of current expenditure. Public investment would subsequently decelerate, while government consumption would increase in 2020.

**Headline inflation was projected to be lower than the previous projection and below the inflation target in 2019 at 0.8 percent** due to lower-than-expected outturns of energy prices and core inflation between June and August 2019. Core inflation would also soften in line with demand-pull inflationary pressures. Energy prices were projected to decline in tandem with global crude oil prices, although fresh food prices were expected to rise in the short term which would support headline inflation to some extent. Nevertheless, **the Committee assessed that headline inflation would increase toward the target in 2020 at 1.0 percent. Core inflation was projected to increase from 0.6 percent in 2019 to 0.9 percent in 2020 in line with the growth outlook and cost-push pressures from the prospects of minimum wage increase next year.**

**The growth and inflation projections were subject to greater downside risks than the assessment in the previous *Monetary Policy Report*** mainly due to external risks. **The possibilities that the Thai economy would underperform the baseline projection** would be due to (1) slower-than-expected trading partner economies and global trade volume as a result of intensifying trade tensions, impinging on Thai merchandise exports more than expected, (2) higher-than-expected increase in crude oil prices due to intensifying tensions in the Middle East, (3) potential implementation delay in government policy and discontinuity from the previous government, and (4) larger-than-expected household income shocks impinging on private consumption. On the other hand, **there were possibilities that the Thai economy would outperform the baseline projection attributable to** (1) higher-than-projected growth of trading partner economies if there were additional government stimulus measures, improvements in trade tensions, and faster-than-expected recovery in the electronics cycle, and (2) higher-than-expected domestic demand as a result of public infrastructure investment and PPP projects, which may boost private investment, as well as the potential implementation of additional government stimulus measures. **Downside risks to headline and core inflation projections remained**, in line with risks to the growth projection.

# 1. Global Economy

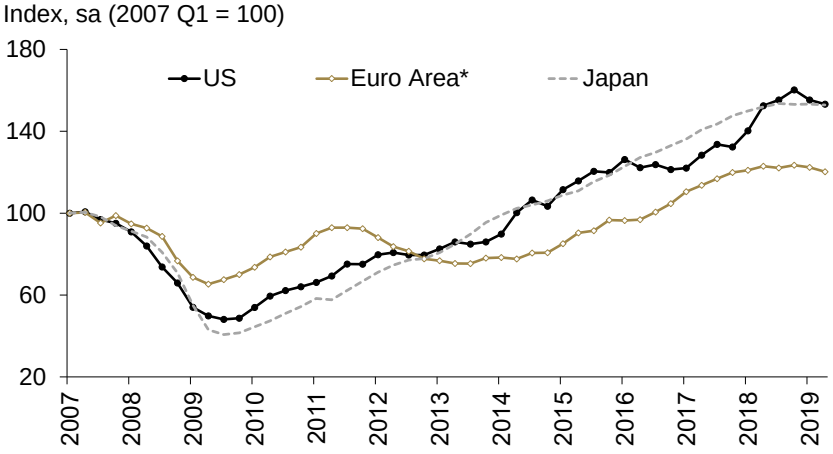
Major advanced economies would slow down in line with production, exports and private investment. However, private consumption continued to be important driver, underpinned by robust employment.

**In the second quarter of 2019, the U.S. and euro area economy expanded at a slower pace than in the previous quarter** on account of moderating exports and private investment. Production also contracted this quarter as a result of intensifying trade tensions, particularly between the U.S. and China, uncertainties surrounding the no-deal Brexit, and prolonged geopolitical issues such as political issues in Italy, which weighed on confidence in the private sector. However, **the Japanese economy** accelerated from the previous quarter driven by improving private consumption, partly due to this year's extended holiday periods.

**In the period ahead, major advanced economies could face risks of a slower economic growth from (1) impacts of trade tensions** on production, exports, and global trade volume, and **(2) impacts of prolonged geopolitical issues** on investor confidence. However, private consumption continued to be important driver given robust labor market, especially employment in the services sector. However, there were signs of somewhat softening labor markets, as indicated by high job vacancy (Chart 1.1). Developments in the labor market thus warranted close monitoring. In this regard, a more accommodative monetary policy stance and greater role of fiscal policy would support growth of major advanced economies, despite external uncertainties in the period ahead.

**Chart 1.1** Labor markets in major advanced economies exhibited modest signs, reflected by high levels of job vacancies

Job vacancy in major advanced economies




Note: * Euro area economy is a GDP-weighted average of Germany, France and Italy.

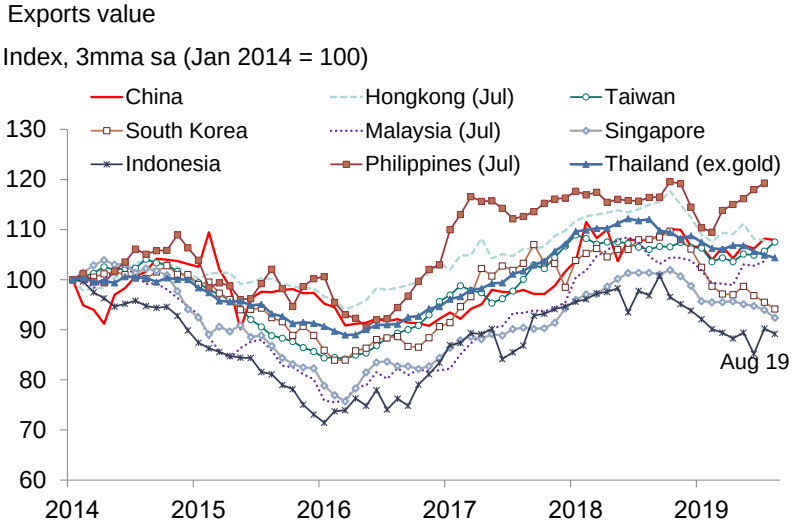
Source: CEIC

Asian economies would expand at a slower pace mainly on account of trade protectionist measures between the U.S. and China.

**In the second quarter of 2019, the Chinese economy expanded at a slower pace than in the previous quarter**, mainly as production and exports were affected by trade protectionist measures. Meanwhile, overall investment improved in tandem with infrastructure investment, partly thanks to government measures to support investment, including the relaxation of regulations to facilitate local governments' issuance of special bonds. In the period ahead, a variety of policy tools to support the domestic economy, such as facilitating local government bond issuances, preparing for proactive measures to stimulate domestic demand, and reducing financing costs to a low level, would help mitigate negative impacts from external factors to some extent and shore up the Chinese economy to achieve the government's economic growth target.

**In the second quarter of 2019, Asian economies (excluding Japan and China) moderated from the previous quarter** as production and exports were affected by trade protectionist measures. There were also country-specific factors in some countries, such as protests in Hong Kong affecting tourist figures, and a delayed approval of the Philippines' government budget for the fiscal year 2019. Looking ahead, Asian merchandise exports were expected to slow down in line with global trade volume and a prolonged down cycle of the electronic products (Chart 1.2). Moreover, businesses postponed their investment as prolonged geopolitical issues in many countries undermined confidence. However, robust labor markets, especially employment in the services sector, and government stimulus measures, such as higher investment in infrastructure projects, welfare subsidies, and minimum wage increases, would help mitigate the impact of the economic slowdown and support domestic demand in the period ahead.

**Chart 1.2** Asian exports contracted in line with global trade slowdown and a prolonged down cycle of electronic products




Source: CEIC

## Growth projections for Thailand's trading partner economies were revised downward, with risks tilting slightly more to the downside.

Going forward, **Thailand's trading partner economies were projected to slow down more than previously assessed in the previous *Monetary Policy Report*.**[1/] This would be due to, first, prolonged and intensifying trade tensions, particularly as the U.S. raised tariffs on Chinese imports worth 250 billion U.S. dollars from a previous rate of 25 to 30 percent, effective on 15 October 2019, and additionally raised tariffs on the remaining Chinese imports worth 300 billion U.S. dollars from a previous rate of 10 to 15 percent, which would be gradually effective on 1 September 2019 and 15 December 2019. Subsequently, China responded by increasing tariffs on U.S. imports worth 75 billion U.S. dollars from 5-25 to 10-35 percent, which would be gradually effective on 1 September 2019 and 15 December 2019. Second, other prolonged geopolitical issues might affect confidence and investment, including protests in Hong Kong, tensions in the Korean Peninsula and the Middle East, as well as disputes between Japan and South Korea. **The Committee thus revised down growth projections for Thailand's trading partners from 3.0 to 2.8 percent in 2019 and from 2.8 to 2.7 percent in 2020**. (Table 1.1)

**Table 1.1** Assumption on trading partner growth

| Annual change (%YoY) | Weight (%) | 2018* | 2019 | 2020 |
|---|---|---|---|---|
| United States | 15.9 | 2.9 | 2.3 (2.5) | 1.7 (1.7) |
| Euro area | 10.3 | 1.9 | 1.2 (1.1) | 1.3 (1.4) |
| Japan | 14.1 | 0.8 | 1.0 (0.9) | 0.2 (0.3) |
| China | 17.1 | 6.6 | 6.1 (6.1) | 5.8 (5.8) |
| Asia (excluding Japan and China)** | 34.2 | 4.0 | 2.9 (3.5) | 3.3 (3.5) |
| Total*** | 100 | 3.5 | 2.8 (3.0) | 2.7 (2.8) |

Note: *Outturn

**Weighted by 7 trading partners' shares in Thailand's exports in 2018, namely, Singapore (5.3%), Hong Kong (7.1%), Malaysia (6.6%), Taiwan (2.2%), Indonesia (5.7%), South Korea (2.8%), and Philippines (4.5%)

***Weighted by 13 trading partners' shares in Thailand's exports in 2018 (including the United Kingdom and Australia)

( ) as reported in Monetary Policy Report, June 2019

**The Committee assessed that there were possibilities that trading partners' growth could underperform the baseline projection in line with slightly greater downside risks.** Such risks included (1) uncertainties pertaining to trade tensions that could be prolonged and thus affect consumption and investment outlook, and the recovery of electronics cycle and exports, and (2) geopolitical risks that could intensify and might lead to

[1/] Growth projections for the euro area and Japan in 2019 were higher than in the previous *Monetary Policy Report* in June 2019, due to revisions of all series, resulting in higher growth after data revision, despite softening economic outlook for these countries.

volatilities in financial and commodity markets. However, risk of no-deal Brexit was close to the previous assessment and continued to prolong.

### Many central banks cut their policy rates, the majority of them shifting toward more dovish stance to shore up the economy

**Monetary policies of major advanced economies' central banks became more accommodative** to cushion against a potential economic slowdown. The Federal Reserve (Fed) cut its policy rate in September 2019 from 2.0-2.25 percent to 1.75-2.0 percent. Another policy rate cut was expected in 2019 with another cut in 2020. The European Central Bank (ECB) announced a package of additional monetary easing measures.[2/] Meanwhile, the Bank of Japan (BOJ) would likely ease monetary policy further should the economy experience a significant slowdown.

**Regional central banks shifted toward a more dovish monetary policy stance** to support weakening economic and inflation outlook. The Reserve Bank of Australia (RBA), the Bank of Korea (BOK), the Reserve Bank of India (RBI), and Bangko Sentral ng Pilipinas (BSP) cut their policy rates by 0.25 percentage point. Bank Indonesia (BI) cut its policy rate three consecutive times, totaling 0.75 percent point, in the third quarter.

### Emerging markets (EMs) experienced net capital outflows on investor concerns over intensifying trade tensions. However, capital flows returned to EMs at the end of the third quarter after those concerns alleviated

In the first half of the third quarter in 2019, investor concerns in the global financial market increased following the Fed's signals on a less dovish monetary policy stance than the markets anticipated, intensifying trade tensions between the U.S. and China, as well as trade tensions among other trading partners such as Japan and South Korea. Moreover, heightened geopolitical risks, especially prolonged and increasingly violent protests in Hong Kong, also dampened investor sentiment. This was reflected by a higher VIX index (Chart 1.3),[3/] a decline in global stock indices, as well as investors' risk-off sentiments and continued portfolio outflows from EMs. In addition to a more accommodative monetary policy of many EM central banks including Thailand, these induced further capital outflows from the bond market. However, at the end of the third quarter, concerns over trade tensions alleviated as the U.S. and China would resume trade negotiations once again in October and postpone the enforcement of import tariffs on certain products prior to the negotiations. Furthermore, central banks of major advanced economies, especially the Fed and ECB, shifted toward a more dovish monetary policy stance. These factors resulted in a decline in the VIX index and capital flows started to return to EMs.

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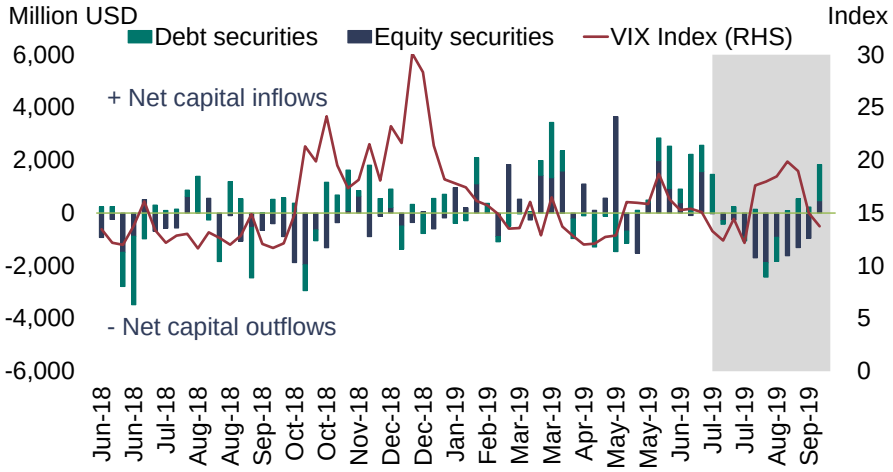
[2/] (1) The ECB cut the deposit facility rate from -0.40 to -0.50 percent, (2) raised the purchase amount under the asset purchase program (APP) to 20 billion euro per month from previously maintaining the reinvestment, (3) amended details and extended the period of the plan to inject liquidity in the financial system through the third targeted longer-term refinancing operations (TLTROs-III) from 2 to 3 years to support a more accommodative monetary policy stance through commercial bank lending channel, and (4) started the two-tier system for commercial banks' reserves, where excess reserves would be partially exempted from the negative deposit facility rate.

[3/] VIX Index is a measure of the stock market volatility implied by S&P 500 Index options

Looking ahead, the global financial markets would remain highly volatile. EMs capital inflows and outflows might fluctuate in tandem with uncertainties of the ongoing trade protectionist measures, advanced economies' monetary policy stance, Brexit developments, as well as geopolitical risks.

**Chart 1.3** EMs experienced capital outflows amid concerns over trade tensions and slowing outlook of the global economy. However, capital inflows returned to EMs following improved sentiments over US-China trade tensions at the end of the third quarter.

Net portfolio inflows to EMs* and VIX index (weekly)




Note: *EMs include Thailand, Indonesia, India, South Africa, and Turkey
Sources: Bloomberg and Institutional Institute of Finance

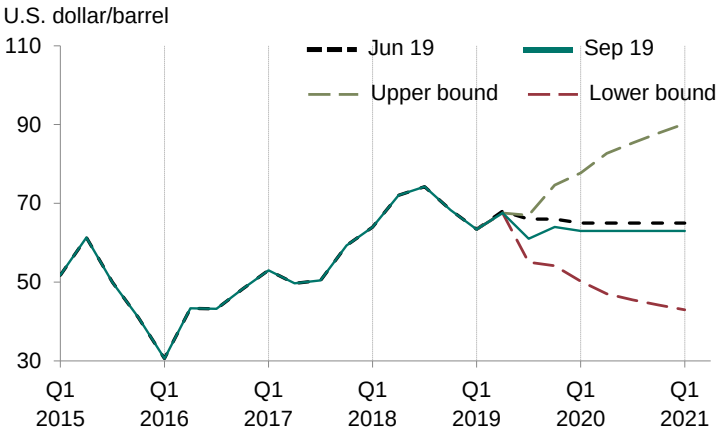
Dubai crude oil prices in the third quarter of 2019 declined from the previous quarter driven by concerns over a softening global economic outlook and lower-than-expected demand for crude oil. At the end of the third quarter, the attacks on Saudi Arabia 's oil processing facilities drove crude oil prices upward.

**Dubai crude oil prices in the third quarter of 2019 decreased from the previous quarter** due to lower crude oil demand than market expected as a result of intensified trade tensions between the U.S. and China. Thus, Dubai crude oil prices decreased to 60.0 U.S. dollars per barrel since the middle of the quarter, despite factors that pushed crude oil prices slightly upward. These factors included (1) a tight crude oil supply due to a continued reduction of the production capacity by OPEC members and allies in accordance with the agreement to cut crude oil production, and (2) attacks on Saudi Arabia's oil processing facilities at the end of the quarter. These heightened tensions in the Middle East and caused the market to expect that Saudi Arabia's crude oil production would be reduced by half.

**The Committee thus revised the projection for Dubai crude oil prices in 2019 and 2020** from 65.7 and 65.0 U.S. dollars per barrel to 64.0 and 63.0 U.S. dollars per barrel, respectively. This downward revision was because the global economy was expected to slow down more than previously assessed following intensified trade tensions. In addition, the International Maritime Organization (IMO)'s enforcement of the new marine fuel regulations from January 2020 onward might lower demand for high-sulfur crude oil, including Dubai crude oil. This factor was expected to have a larger impact on crude oil prices during the

projection period than the attacks on Saudi Arabia's oil processing facilities, whose impact was expected to be resolved in the short run. Moreover, **risks to the projection of Dubai crude oil prices were revised to be in balance**, from tilting downward in the previous *Monetary Policy Report*. This was as there were higher possibilities of higher-than-baseline crude oil prices given potentially intensifying geopolitical issues, especially in the Middle East, after the attacks on Saudi Arabia's oil processing facilities.

**Chart 1.4** Dubai crude oil prices in 2019Q3 decreased from the previous quarter due to concerns over global economic slowdown and lower crude oil demand than market expected



# 2. The Thai Economy

## 2.1 Recent developments

The Thai economy expanded at a slower pace in the second quarter of 2019 due to slowdown in both domestic and external demand.

In **the second quarter of 2019**, the Thai economy expanded 2.3 percent from the same period last year. Growth moderated from the previous quarter and was lower than previously assessed in the previous *Monetary Policy Report*, due to domestic and external demand slowdown. Private consumption moderated mainly due to spending on durable goods and services. Moreover, factors supporting overall purchasing power weakened, as reflected by declining consumer confidence and household income slowdown consistent with lower employment. However, consumption was supported by additional government measures aiding lower income groups through the social welfare card scheme, which shored up household purchasing power. Private investment slowed down on account of both residential construction and machinery and equipment. Public spending growth decelerated mainly due to personnel compensation expenditure, while public investment growth resumed thanks to accelerated construction investment of the central government. With regard to external demand, merchandise exports continued to contract across most product categories due to the slowdown in trading partner economies, trade protectionism between the U.S. and China, as well as a prolonged down cycle of the electronic products. Manufacturing exports contracted, including electronic components and devices, automobile, and petroleum-related products. Exports of agricultural products continued to contract as prices of Thai rice exports in U.S. dollar terms were higher than those of competitors, while export volume of rubber was affected by the Agreed Export Tonnage Scheme (AETS) of the International Tripartite Rubber Council (IRTC).[4/] Exports of services continued to contract as tourism receipts declined in line with moderation in foreign tourist figures. Overall, the Thai economy expanded 0.6 percent in the second quarter of 2019 from the previous quarter, seasonally adjusted.

In **the third quarter of 2019,** most recent economic indicators reflected that the Thai economy would expand at a slower rate relative to the previous quarter. Private consumption grew at a low rate relative to the first half of the year, particular durable goods, in line with a continued decline in consumer confidence. In addition, employment of both non-farm and farm households dropped, partly as a result of export contraction in the previous period. However, the government increased extra budget for the social welfare card scheme, which was expected to shore up household purchasing power to some extent. Private investment slowed down in line with declining investment in residential construction. However, investment in machinery and equipment of some industries expanded thanks to the relocation of production base to Thailand to mitigate the impact of trade protectionism. Government spending, excluding transfers, grew on the back of both current and capital expenditures. Meanwhile, public investment expanded mainly due to the Department of Highways' disbursement. Merchandise exports continued to contract owing to a slowdown in the global economy and global trade volume, as well as a prolonged down cycle of the electronic products. However, exports of some product categories improved, such as home electrical appliances, thanks to trade diversion effect after having been affected by the impact of trade

[4/] Members of the International Tripartite Rubber Council (IRTC) include Thailand, Indonesia, and Malaysia.

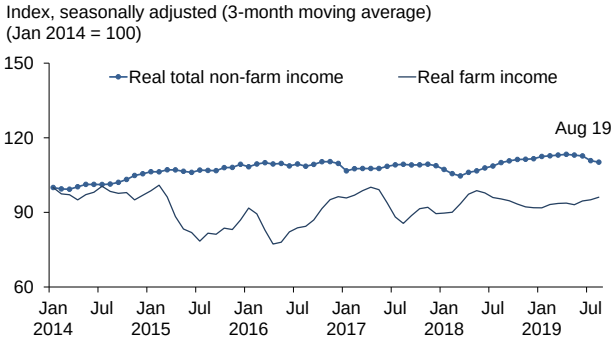
protectionism. Moreover, some industries that relocated their production base to Thailand in earlier periods started to export their products, including hard-disk drives. Growth in the exports of services resumed slightly on account of an increase in foreign tourist figures, especially Chinese and Indian tourists, partly due to a low base effect in the previous year from the Phuket tour boat sinking incident. In addition, the extended exemption of the visa-on-arrival (VOA) fees that would have ended in October 2019 would boost foreign tourist figures to some extent.

## Overall purchasing power slowed down, mainly on account of contraction in employment and income of non-farm households, with signs of employment reduction across all industries. Purchasing power of farm households affected by climate variability warranted monitoring.

Moderating economic growth resulted in a slowdown of overall household purchasing power, consistent with employment reduction (Charts 2.1 and 2.2). Employment and income of **non-farm households** contracted from the same previous last year across all sectors. In particular, normal-hour and over-time employment fell in export-related manufacturing sectors, including food, electronics, electrical appliances, and motor vehicles. Meanwhile, employment in tourism-related service sector declined in line with a slowdown in foreign tourist figures.

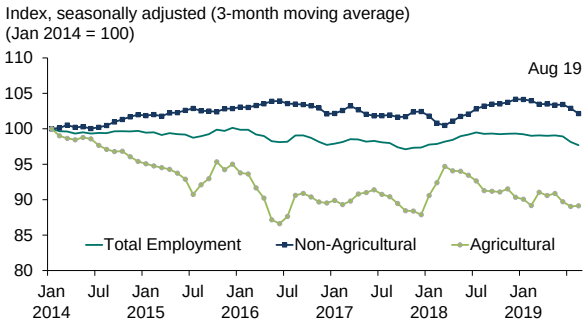
Employment of **farm households** contracted, partly as a result of drought in the first half of the year. However, farm income improved slightly supported by an increase in prices, especially sticky rice prices due to lower market output as a result of drought. Prices of rubber and palm oil increased slightly from the previous year, while agricultural output was stable. Going forward, there remained a need to closely monitor farm household income and purchasing power. This was due to recent climate variability that continued until the third quarter of this year, which caused drought and floods in agricultural areas, as well as devastated crop production.

**Chart 2.1** Overall purchasing power slowed down
Household income indicators




Note: All indicators are calculated using real-term values to extract the effect of price factors.
Sources: Bank of Thailand, Office of Agricultural Economics, National Statistical Office, Ministry of Commerce, calculations by Bank of Thailand

**Chart 2.2** Overall employment contracted. Employment of non-agricultural households fell mainly in export-related industries.
Employment indicators



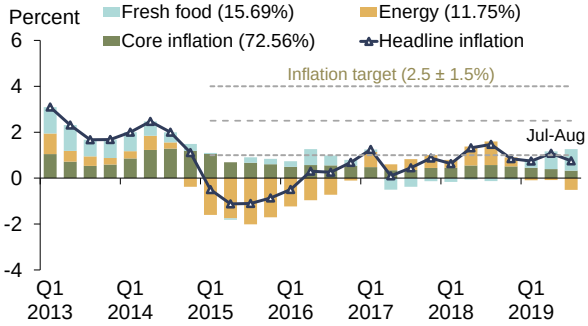

Source: National Statistical Office

## Headline inflation declined in line with energy prices and core inflation.

**Headline inflation** averaged at 0.75 percent in the first two months of the third quarter of 2019, down from 1.08 percent in the previous quarter (Chart 2.3). The decrease was due to falling energy prices, especially domestic retail oil prices which fell in tandem with global crude oil prices, as well as the high-base effect. Meanwhile, fresh food prices continued to increase following higher prices of rice and fruit due to lower production.

**Chart 2.3** Headline inflation declined from the previous quarter mainly due to energy prices

Headline inflation and inflation target



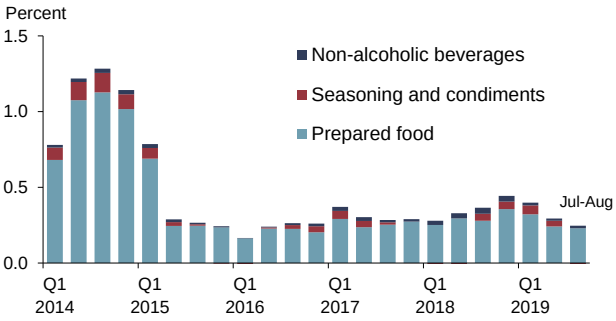

Note: Headline inflation for the first 8 months of 2019 recorded 0.87 percent
( ) denotes share in inflation baskets
Source: Ministry of Commerce, calculations by Bank of Thailand

**Core inflation** averaged at 0.45 percent for the first two months of the third quarter of 2019, down from 0.54 percent in the previous quarter. Food-in-core inflation fell (Chart 2.4) mainly in line with prices of seasoning and condiments. Prepared food prices dropped slightly, reflecting low pass-through from higher costs of fresh food to prepared food prices. Meanwhile, core inflation in the non-food categories decreased slightly (Chart 2.5) in tandem with prices of medical and personal care, especially a fall in prices of personal care products and subdued house rents. In other categories, prices slowly increased following economic slowdown, and was influenced by changes in structural factors including the expansion of e-commerce, rising price competition, and lower production costs from higher production efficiency (Table 2.1).

**Short-term (one-year ahead) inflation expectations** according to the survey of businesses in August 2019 stood at 1.8 percent, a slight decline from 1.9 percent in the previous quarter. Inflation expectations of professional forecasters in June 2019 stood at 1.3 percent, down from 1.5 percent in the previous quarter. Meanwhile, **long-term (five-year ahead) inflation expectations** according to the survey of professional forecasters in June 2019 stood at 1.8 percent, unchanged from the previous survey (October 2018). In addition, inflation expectations based on macro-finance term structure model with bond yields and macroeconomic data stabilized at 1.6 percent in the second quarter of 2019.

**Chart 2.4** Core inflation in the food category (28% of core inflation) declined mainly in line with seasoning and condiment prices

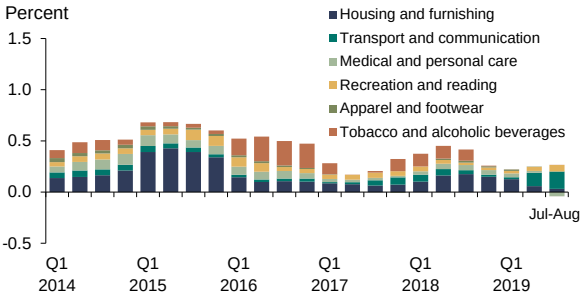
Contribution* to core inflation in the food category




Note: *Contributions to core inflation decompose core inflation into an inflation rate of each component within the core inflation basket, weighted by its corresponding share in the basket.
Source: Ministry of Commerce, calculations by Bank of Thailand

**Chart 2.5** Core inflation in the non-food category (72% of core inflation) decreased in line with personal care price and house rent

Contribution* to core inflation in the non-food category




Note: *Contributions to core inflation decompose core inflation into an inflation rate of each component within the core inflation basket, weighted by its corresponding share in the basket.
Source: Ministry of Commerce, calculations by Bank of Thailand

**Table 2.1** Inflation

| Annual percentage change | 2017 | 2018 | | | | 2019 | | |
|---|---|---|---|---|---|---|---|---|
| | | Q1 | Q2 | Q3 | Q4 | Q1 | Q2 | Jul - Aug |
| Headline Consumer Price Index (Headline CPI) | 1.07 | 0.64 | 1.31 | 1.47 | 0.84 | 0.74 | 1.08 | 0.75 |
| Core Consumer Price Index (Core CPI) | 0.71 | 0.61 | 0.76 | 0.78 | 0.71 | 0.62 | 0.54 | 0.45 |
| Raw food | -0.63 | -1.04 | -0.35 | -0.82 | -0.35 | 2.50 | 4.98 | 6.12 |
| Energy | 5.66 | 3.01 | 7.30 | 9.11 | 3.39 | -0.79 | -0.70 | -4.24 |

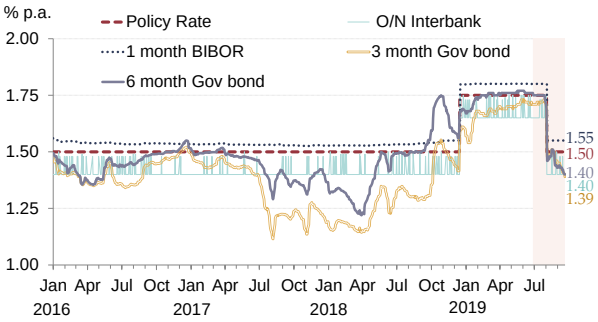
Source: Bureau of Trade and Economic Indices, Ministry of Commerce

## Short-term money market rates, short-term government bond yields, and some types of bank lending rates declined in tandem with the policy rate. Long-term government bond yields fell rapidly driven by both domestic and external factors.

In the third quarter of 2019, **short-term money market rates and short-term government bond yields** continuously declined in line with the policy rate and due to an increase in mutual funds' short-term bond holdings after the withholding tax imposed on fixed-income fund became effective on August 20, 2019 (Chart 2.6). **Medium- and long-term government bond yields** (Chart 2.7) continuously declined contributed by both domestic and external factors, including (1) policy rate cut by 0.25 percent to 1.5 percent on August 7, 2019, (2) accelerated purchases of long-term government bonds by mutual funds before the withholding tax levied on bond investment became effective, while some other investors changed strategies to increase long-term bond holdings, and (3) risk-off sentiment in global financial markets owing to trade tensions, global economic slowdown, as well as the likelihood of additional policy rate cuts by central banks of advanced economies. Nevertheless, long-term government bond yields fell faster than short-term bonds, resulting in flatter yield curve relative to the end of previous quarter.

**Chart 2.6** Short-term money market rates and short-term government bond yields declined in line with the policy rate

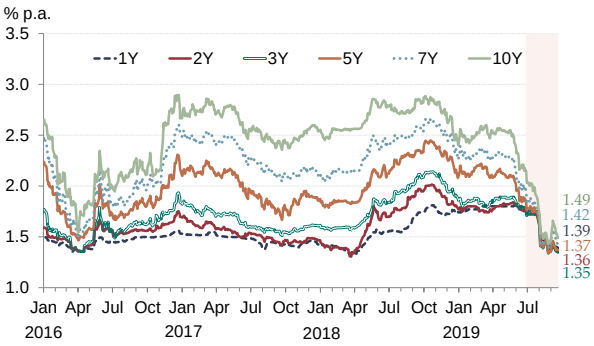
Short-term rates in financial markets




Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA) (data as of 24 Sep 2019)

**Chart 2.7** Long-term government bond yields fell rapidly due to both domestic and external factors

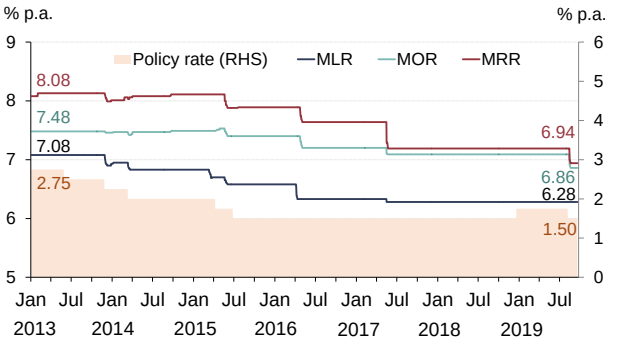
Government bond yields




Source: Thai Bond Market Association (Thai BMA) (data as of 24 Sep 2019)

**Overall corporate bond yields** declined in the third quarter of 2019 in tandem with the policy rate. Similarly, **costs of financing through commercial banks** became more accommodative. Most Thai commercial banks reduced their minimum retail rates (MRR) and minimum over draft rates (MOR) in August 2019 (Chart 2.8). Meanwhile, the minimum loan rate (MLR) remained unchanged. The new loan rate (NLR)[5/] was stable at 4 percent in July 2019 relative to the previous quarter end.

**Chart 2.8** Commercial banks reduced MRR and MOR in tandem with the policy rate, while MLR remained unchanged.

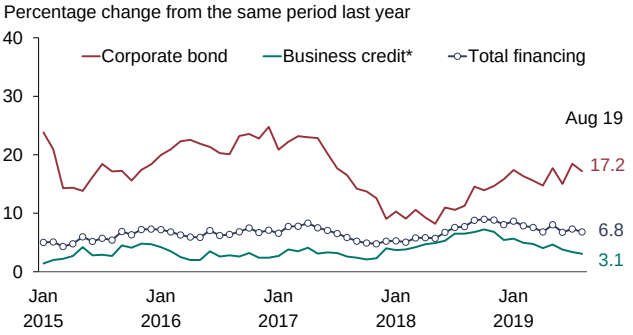



Note: Average of 5 largest commercial banks' lending rates
Source: Bank of Thailand (data as of 24 Sep 2019)

## Financial conditions remained accommodative for business financing, especially in the bond market.

**Net issuance of corporate bonds** in July and August 2019 continued to expand from the previous quarter, driven by businesses related to real estate. Most funding were for merger and acquisitoin (M&A) purposes and to refinance existing debt due to be repaid in the third quarter. These activities caused the outstanding stock of corporate bonds in August 2019 to expand by 17.2 percent from the same period last year (Chart 2.9). **Equity financing** increased from businesses in information and communication technology, real estate and construction, as well as services, mostly for investment in M&A.

**Chart 2.9** Overall business financing decelerated slightly
Growth of outstanding corporate bonds and business credit




Note: *Business credit covers lending activities of Other Depository Corporation (ODCs)
Sources: Thai Bond Market Association (Thai BMA) and Bank of Thailand

Nevertheless, **private credit**[6/] slowed down to 4.1 percent (Chart 2.10) due to both business and household credit. Business credit slowed down mainly in manufacturing, commercial, and service sectors in line with private consumption and investment. Household credit moderated in line with mortgage loans which had been front-loaded before the implementation of the revised loan-to-value (LTV) ratio measure in April 2019. Meanwhile, auto loans started to decelerate in line with car sales.

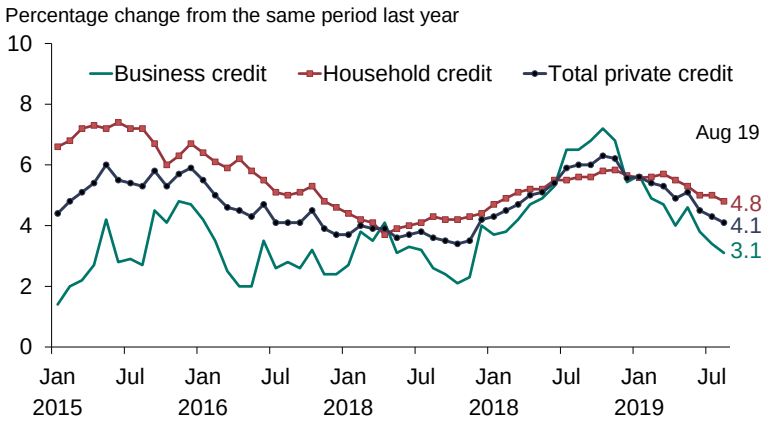
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5/ NLR is the weighted average of interest rates on new loan contracts extended by 14 Thai commercial banks each month. The loan contracts exclude consumer loans, credit card loans, repurchase agreements, bank guarantees, as well as loans extended to financial intermediaries, the public sector and non-residents. The dataset covers loans with value of 20 million baht or higher for all loan types, purposes and maturities, and includes both secured and unsecured loans. Moreover, interest rates used in the calculation refer to the mid-rate between the lowest and the highest rates in each loan contract.

6/ Outstanding credit of other depository corporations (ODCs), namely commercial banks, specialized financial institutions, finance companies, savings cooperatives, and market mutual funds.

**Chart 2.10** Private credit decelerated in both businesses and households

Growth of private credit



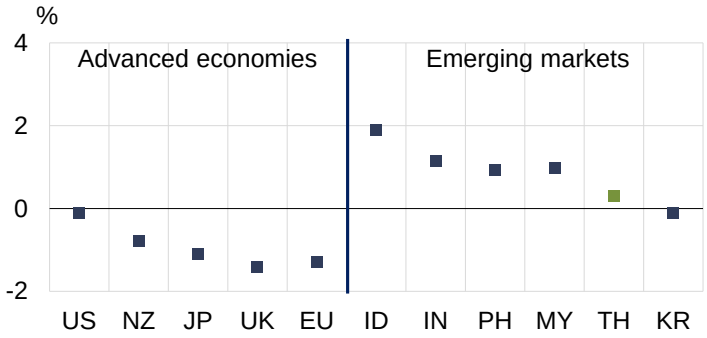

Note: 1) Private credit includes credit to other depository corporations (ODCs) namely commercial banks, specialized financial institutions, finance companies, saving cooperatives, and money market mutual funds
2) The data for outstanding ODCs credit to businesses and households since January 2015 have been revised in line with improvements in the processing system for more accurate and comprehensive ODCs credit data

Source: Bank of Thailand

**Overall financial conditions remained accommodative** as reflected by a decline in real policy interest rate from the previous quarter after the policy rate cut. The rate was still low relative to other emerging markets (Chart 2.11), indicating Thailand's relatively accommodative financial conditions. However, in the third quarter of 2019[7/], financial institutions were expected to tighten credit standards for both loans to businesses and households, particularly (1) loans extended to export-related SMEs, which may be affected by external demand slowdown, and (2) credit card loans, as borrowers' credit quality had still not improved.

**Chart 2.11** Thailand's real policy rate* decreased due to the policy rate cut and remained at a low level compared with emerging market (EM) economies

Real policy rates*




Note: *Calculated from policy rate subtracted by one-year-ahead inflation expectation according to a survey by Consensus Economics (data as of 9 September 2019)

Sources: Bloomberg, Consensus Economics, calculation by Bank of Thailand

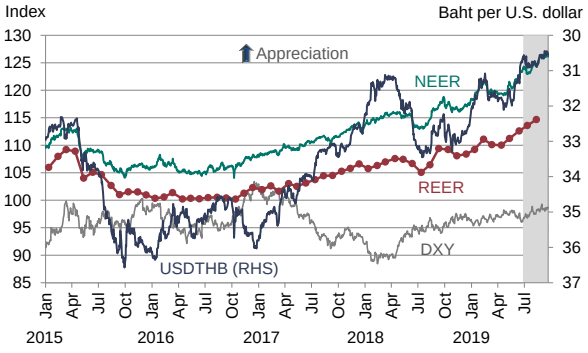
[7/] Results of financial institutions' survey as published in the Report on Credit Conditions Q2/2019 and Outlook for Q3/2019

## The baht appreciated against the U.S. dollar, while the nominal effective exchange rate (NEER) index continued to appreciate.

In the third quarter of 2019, **the baht appreciated against the U.S. dollar** relative to the end of the previous quarter (Chart 2.12). At the beginning of the quarter, the baht depreciated in line with U.S. dollar appreciation following better-than-expected U.S. economic data outturns and in response to measures enhancing the surveillance of short-term capital inflows by the BOT[8/]. The measures partly helped to reduce channels for speculation on baht appreciation and weakened the baht to some extent. However, since early August, the baht appreciated against the U.S. dollar once again, driven by concerns over global economic slowdown and intensifying and prolonged trade tensions, particularly after the U.S. announced additional tariffs on Chinese imports and designated China as a currency manipulator. Such concerns raised the prices of safe-haven assets including gold, the yen, and the baht. On August 7, 2019, the policy rate cut by 0.25 percent to 1.5 percent helped to slow down the pace of the baht appreciation to some extent. At the end of the quarter, the baht strengthened in line with regional currencies following (1) less intensified trade tensions, (2) lower likelihood of no-deal Brexit, and (3) more accommodative stance of advanced economy central banks. As of September 24, 2019, the baht closed at 30.55 baht to U.S. dollar, a 0.4 percent appreciation from the end of previous quarter.

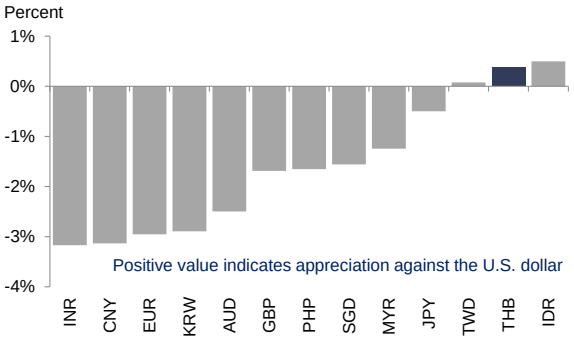
**The nominal effective exchange rate (NEER)** index as of September 24, 2019 stood at 126.1, an appreciation of 2 percent from the end of the previous quarter, consistent with baht appreciation against most trading partner currencies except the rupiah (Chart 2.13). The preliminary **real effective exchange rate (REER)** at the end of August 2019 rose about 2 percent from the previous quarter end. Going forward, exchange rates would likely remain volatile due to uncertainties surrounding trade tensions, monetary and fiscal policies of major advanced economies and regional economies, as well as geopolitical risks.

**Chart 2.12** The baht appreciated against the U.S. dollar
USDTHB, NEER, DXY




Source: Bank of Thailand and Reuters (data as of 24 September 2019)

**Chart 2.13** The baht appreciated against the U.S. dollar, while regional currencies depreciated (24 Sep 19 compared to the end of Jun 19)




Source: Bank of Thailand and Reuters (data as of 24 September 2019)

8/ On July 12, 2019, BOT announced measures to enhance surveillance of short-term capital inflows including (1) a reduction of limit on outstanding balance of Non-Resident Baht Account (NRBA) and Non-Resident Baht Account for Securities (NRBS) from 300 to 200 million baht per non-resident and (2) a requirement to report ultimate beneficiary owner (UBO) to enhance BOT's surveillance of non-resident investment behaviors.

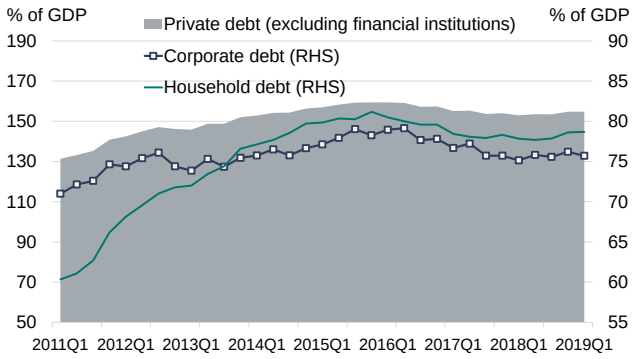
There remained pockets of risks in the financial system that could pose vulnerabilities to financial stability in the future. These risks included (1) weakening ability of households to cope with negative income shocks, (2) rising risks of oversupply in the real estate market following domestic and foreign demand slowdown, and (3) the search-for-yield behavior which could lead to underpricing of risks, particularly among saving cooperatives and large corporates.

Thailand's financial stability remained sound overall, particularly with regard to external stability, as indicated by the country's high level of international reserves and sustained current account surpluses. Meanwhile, the low external debt to GDP ratio provided cushion for the Thai economy against risks of volatile global financial markets. Financial institutions maintained robust financial positions, reflected by high levels of capital buffers, providing cushion against risks of deteriorating credit quality. Nevertheless, there remained other risks that warranted continued monitoring:

**1. Weakening ability of households to cope with negative income shocks.** Household debt to GDP ratio in the first quarter of 2019 increased from the previous quarter to 78.7 percent (Chart 2.14). In addition, household income was expected to slow down in line with the economy. These raised household vulnerability and could lead to defaults. Although the NPL ratio of consumer loans in the second quarter of 2019 was stable at 2.7 percent, there were signs of increasing special mention loan (SM) ratio, which reflected the inability to service consumer loans for 1-3 months. The SM ratio increased across all loan types, particularly auto loans. Going forward, some household loans were expected to slow down, especially mortgage loans due to frontloading effect prior to the implementation of the revised LTV measure. Auto loans tended to moderate in line with car and motorcycle sales. However, certain types of loans which households use for enhancing liquidity were expected to accelerate, such as home-for-cash loans, car-for-cash loans, as well as credit card and cash card loans. Nevertheless, issues that warranted close monitoring in the period ahead included (1) severity and length of negative economic shocks, such as impacts from trade protectionist measures between the U.S. and China, which would affect economic growth and household income, and (2) households' ability to maintain liquidity during the economic downturn.

**Chart 2.14** The ratio of household debt to GDP exhibited a rising sign

Private debt to GDP




Source: Bank of Thailand

Financial position of businesses became more vulnerable, especially small enterprises. This was reflected by sustained negative operating profit margin (OPM) and interest coverage ratio (ICR) of some small businesses[9/]. Furthermore, the NPL ratio of commercial banks' SME loans in the second quarter of 2019 remained elevated at 4.5 percent.[10/]

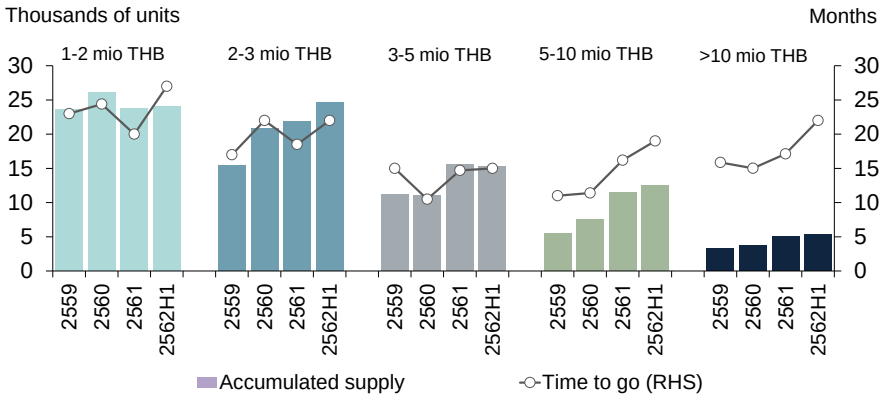
9/ As reflected by the ICR and OPM of 25th-percentile small-sized listed companies which have been in the negative territory

10/ NPL ratio of SME loans from 2013 to date averaged at 3.9 percent.

**2. Risks of oversupply in the real estate market increased.** This was reflected by rising oversupply and longer 'time to go' (in Bangkok and vicinities), particularly for condominium where unsold units increased 8.5 percent from the end of 2018 and time-to-go rose from 18 to 21 months. When considering prices and areas, condominiums with higher unsold units and longer 'time to go' were those above 5 million baht (Chart 2.15) and those near the MRT Blue Line (Bang Sue – Tha Phra) (Chart 2.16), following increased number of project launches. This was reflected by expanding number of newly open residential units (in Bangkok and vicinities) during the first seven months of 2019, which grew 19.5 percent from the same period last year. Moreover, there remained a large number of new projects in developers' pipeline[11/], despite the number falling from announcements at the beginning of the year. These suggested that launches of condominium projects during the second half of 2019 may not be consistent with domestic and external demand slowdown, as well as with the revised LTV measure. Supply that did not match demand in the real estate market might increase liquidity risks of developers going forward. (Box: *Adjustments in the real estate sector following the revision of the loan-to-value (LTV) measure*)

**Chart 2.15** The oversupply and 'time-to-go' of condominium in Bangkok and vicinities increased, especially those with value exceeding 5 million baht

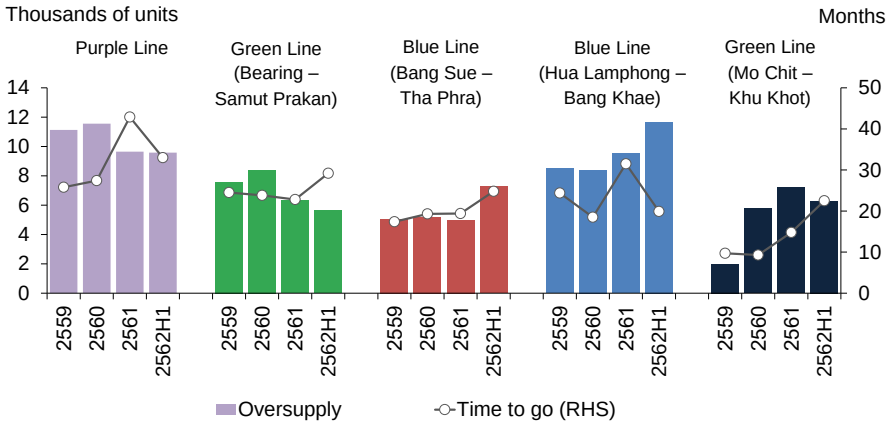
Condominium inventory in Bangkok and vicinity and 'Time to go' by price ranges




Source: AREA and calculated by Bank of Thailand

**Chart 2.16** The oversupply and 'time-to-go' of condominium around the MRT Blue Line (Bang Sue – Tha Phra) increased

Accumulated condominium oversupply and 'time-to-go', by skytrain lines



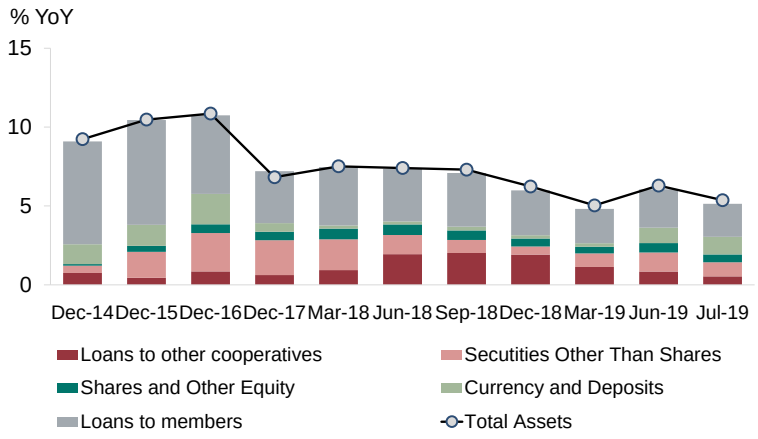

Source: AREA, calculations by the Bank of Thailand

11/ Data from 10 large developers listed in the Stock Exchange of Thailand

**3. Continued search-for-yield behavior under prolonged low interest rates could lead to underpricing of risks.** Although systemic risks to financial stability remained stable overall, there remained risks that warranted monitoring. First, search-for-yield behavior of saving cooperatives continued, despite the slower expansion in assets in line with deposits and equities. Saving cooperatives increasingly raised funds through deposits and reduced their borrowing (Chart 2.17) after regulatory authorities cooperated in revising regulations to enhance supervisory standards. However, search-for-yield behavior warranted monitoring, particularly the rise in corporate bond holdings of some large saving cooperatives. Moreover, overall systemic risk in the saving cooperatives system increased from higher borrowings between saving cooperatives. Second, some large corporates accelerated their fundraising under prolonged low interest rates. These corporates had high interconnectedness with the financial system, especially through bond and equity financing, which could lead to concentration risks in the financial markets. Third, offshore investment through foreign investment funds (FIF) continued to expand. The risk of FIF investments being concentrated in only a few countries warranted monitoring.

**Chart 2.17** Assets of saving cooperatives expanded at a slower rate, but there remained pressure to search for higher yield

Contribution to growth of savings cooperatives' assets




Note: Saving cooperatives were subjected to tighter regulation by government since H2/2017

Source: Cooperative Auditing Department, calculations by Bank of Thailand

## Adjustments in the real estate sector following the revision of the loan-to-value (LTV) measure

During 2017 – 2018, there was a build-up of vulnerabilities the real estate sector, particularly in the mortgage loan market. This was due to higher competition in loan extension among financial institutions, resulting in deteriorating credit underwriting standards. In some cases, there were no requirements for borrowers to make down payment or to have savings before acquiring loans. In other cases, financial institutions extended loans with value exceeding collateral value. Moreover, there were speculative signs from real estate purchases, expecting returns without real demand in the property. Property was instead rented or invested with an expectation of future price increases. Speculative demand from these buyers caused real estate prices to rise faster than growth of average household income. Thus, buyers with real demand had to buy real estate at higher prices. Meanwhile, real estate developers continued to raise funds in order to expand businesses. In addition, excessive competition among financial institutions and developers might accelerate speculation in the real estate sector and increase risks to macroeconomic and financial stability in the future.

The Bank of Thailand (BOT) recognized the importance of such issue since the real estate sector was one of the root causes of financial crises around the world. The BOT thus implemented a preventive measure to address systemic risks through the revision of the loan-to-value (LTV) measure. Key revision[12/] included a more tightened ceiling of the loan-to-value (LTV) ratio for second and subsequent mortgage contracts, or for residential units with value exceeding 10 million baht. The revised measure became effective on 1 April 2019. This measure aims to enhance credit underwriting standards of financial institutions in order to emphasize lending to residential buyers with real demand, as well as to address over-indebtedness issue and speculation in the real estate market. Therefore, buyers with real demand would be able to buy residential units at more affordable prices.

After the implementation of LTV measure, the real estate market was affected from a slowdown in external demand and softening domestic economy. The BOT had been closely monitoring adjustments of related parties and found that households, financial institutions, and real estate developers had adapted themselves in accordance with the objectives of this measure as follows.

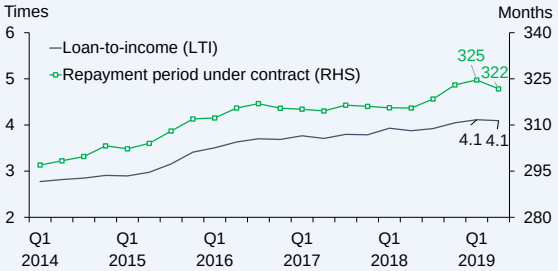
**(1) Credit underwriting standards among financial institutions became more tightened, particularly mortgage contracts on second and subsequent residential units acquired by the same purchaser. However, there was a limited impact on first contract borrowers.** This was reflected by a decline in average LTV ratio of mortgage contracts on second and subsequent residential units from 87.8 percent in the first quarter to 83.1 percent in the second quarter of 2019, while the ratio of first mortgage contracts remained at 88.5 percent (Chart 1). Moreover, the repayment period started to taper off. The average loan-to-income ratio was largely stable. (Chart 2)

**Chart 1** Average loan-to-value (LTV) ratio in the banking system




Note: Calculated from the database on loans that use new housing as collateral
Source: Bank of Thailand

**Chart 2** Average loan-to-income (LTI) ratio and repayment period under contract of the banking system




Note: Calculated from the database on loans that use new housing as collateral
Source: Bank of Thailand

[12/] For more detail, https://www.bot.or.th/Thai/FinancialInstitutions/Publications/Pages/HousingLoan_Notification.aspx

**(2) Speculation in the real estate market was subdued,** as reflected by a contraction in a number of new mortgage contracts during the first half of 2019. In particular, a number of new mortgage loans on second and subsequent contracts, especially for high-rise residential units, fell 25.0 percent. (Table 1) However, a number of first mortgage contracts still expanded 14.0 percent. Overall, a number of new mortgage contracts decelerated from a high expansion of 27.9 percent in the first quarter, before the LTV measure took effect, to only 2.4 percent in the second quarter following a sharp decline of 39.0 percent for mortgage loans on second and subsequent contracts.

**(3) Residential prices moderated (Chart 3), which partly helped residential buyers with real demand to purchase at a more reasonable price.** Credit underwriting standards among financial institutions were more tightened, resulting in demand slowdown. This could partly slow the pace of residential price increases, particularly condominium, whose prices had accelerated significantly in the previous period due to speculation.

Table 1 Growth of new mortgage loan accounts extended by financial institutions

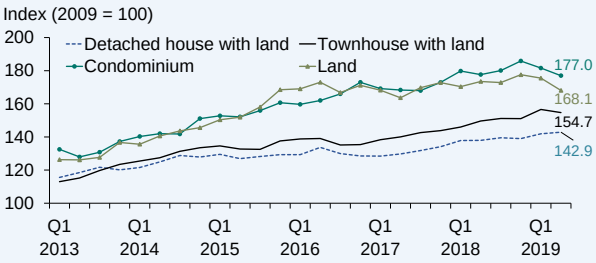
%YoY

| | New mortgage loan accounts |
|---|---|
| Jan – Jun 19 | 14.2 |
| 1st contract* | 14.0 |
| - low-rise* | 17.9 |
| - high-rise* | 5.8 |
| 2nd and subsequent contract* | -13.0 |
| - low-rise* | 3.7 |
| - high-rise* | -25.0 |
| Jan – Mar 19 (Q1/2019) | 27.9 |
| Apr – Jun 19 (Q2/2019) | 2.4 |
| 1st contract* | -2.9 |
| 2nd and subsequent contract | -39.0 |

Note: Calculated from the database on loans that use new housing as collateral

Source: Bank of Thailand

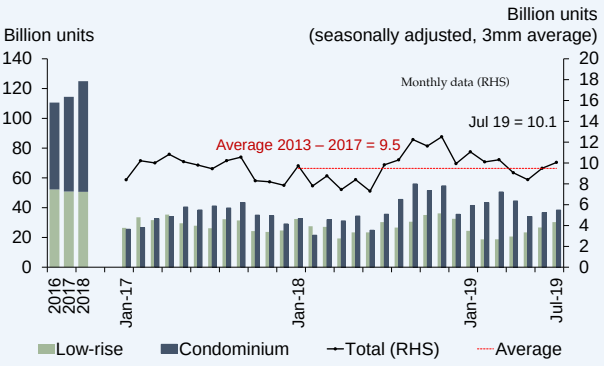
**Chart 3** Real estate price indices for real estates in Bangkok and the vicinities




Note: Indices are calculated based on database on commercial banks' new loans

Source: Bank of Thailand

**Chart 4** New residential units launched in Bangkok and vicinities




Source: AREA, calculation by Bank of Thailand

**(4) Developers adjusted in response to demand slowdown** by focusing on sales promotion strategies for completed projects in order to release high stock of unsold units, as well as focusing on launches of low-rise residential projects which had shorter time-to-go than condominium projects to realize income faster. However, there remained a need to closely monitor issues on an oversupply in the period ahead despite some adjustments among developers. This was because developers listed in the stock exchange set a target to launch a large number of new projects in 2019, both low-rise and high-rise projects, although the figures were somewhat lower than original target set at the beginning of the year. Also, funding through pre-finance and the bond market was expected to rise. Nevertheless, if sales from new project launches were not as expected by developers, especially given the softening outlook of both domestic and external demand, this might increase risks of oversupply and pose liquidity risk for developers going forward. (Chart 4)

Following the implementation of the LTV measure, the BOT had been monitoring the situation and receiving opinions and suggestions from all related parties. The BOT considered a relaxation of this measure in relation to co-borrowers for those who were not registered as owners of the property. This was aimed to mitigate the impact on co-borrowers, most of whom were not high income earners

and could become potential demand for residential units[13/], to acquire mortgage loans in a more appropriate practice. That is, if borrowers were not registered as owners of the property, they would be treated as if they were not borrowers on that contract. This was because they have no real intention to buy the residential property. Co-borrowers might only want to support main borrowers, who were mostly their family members, to acquire mortgage loans from financial institutions. Nevertheless, there remained a need to continue monitoring impacts of the revised LTV measure, as well as adjustments in the real estate sector and oversupply going forward.

[13/] Co-borrowers applying for new mortgage loans under the first contract accounts for 26 percent, most of which are contracts for low-rise houses with value around 2-5 million baht.

## 2.2 Outlook for the Thai Economy

**The Committee assessed that the Thai economy would expand at a slower pace and below potential. The growth forecasts for 2019 and 2020 were 2.8 and 3.3 percent respectively, lower than the assessment in the previous *Monetary Policy Report*** due to slowdown in both external and domestic demand. Exports of goods and services were expected to exhibit slower growth than previously estimated due to further slowdown in global trade volume and trading partner economies. In addition, falling merchandise exports began to impact private consumption and private investment. Meanwhile, public expenditure would grow at a slower rate owing partly to delays in state-owned enterprise investment projects. The annual averages of headline inflation were projected to be lower particularly in 2019, due to falling energy prices in line with global economic slowdown, as well as lower core inflation in line with demand-pull pressures. Nonetheless, headline inflation was projected to increase toward the inflation target in 2020 in tandem with the economic outlook and cost-push pressures from the prospects of minimum wage increase next year.

**Table 2.2** Forecast summary

| Percent | 2018* | 2019 | 2020 |
|---|---|---|---|
| GDP growth | 4.1 | 2.8 (3.3) | 3.3 (3.7) |
| Headline inflation | 1.1 | 0.8 (1.0) | 1.0 (1.0) |
| Core inflation | 0.7 | 0.6 (0.7) | 0.9 (0.9) |

Note: * Outturn
( ) Monetary Policy Report June 2019
Sources: NESDB, Ministry of Commerce, Bank of Thailand's estimates

### Summary of the key forecast assumptions

- Trading partner economies were projected to slow down throughout the forecast horizon due to (1) outturns in the second quarter and indicators in the third quarter of 2019 of many economies being lower-than-expected and (2) intensifying and prolonged international trade tensions, as well as escalating geopolitical risks. Such risks posed greater impacts on global trade volume, investment and production in major advanced economies and would have spillover effects on regional economies.
- The federal funds rate declined in September 2019 to 1.75-2.00 percent. One additional cut was expected this year. In total, there would be three rate cuts in 2019 and one more rate cut in 2020, compared with the previous assessment of two policy rate cuts in 2019 and one additional rate cut in 2020.
- Regional currencies (excluding the Chinese yuan) in the third quarter of 2019 depreciated due to intensifying trade tensions after the U.S. and China mutually raised import tariffs. Nonetheless, regional currencies would gradually appreciate in the period ahead in tandem with the U.S. policy rate.
- Dubai crude oil prices were revised down throughout the forecast horizon mainly due to the global economic outlook, despite increased prices in the short term from reduced supply following an attack on oil facilities in Saudi Arabia.

- Farm income, excluding government support measures, was expected to expand in 2019 at a similar pace as previously estimated. The drought situation in the Northeast would lead to slower agricultural output growth and push up prices, especially glutinous rice and jasmine rice. In 2020, farm income was projected to decline due to the high base effect of agricultural prices this year. However, government support measures, including the project to mitigate the impact of delayed rainfall in 2019 and income guarantee scheme for farmers, would partly shore up farm income.
- Public spending at current prices were revised down throughout the forecast horizon mainly due to delays in state-owned enterprise investment. Moreover, there was a change in budget structure of the Annual Budget Expenditure Act, B.E. 2563 (A.D. 2020) that led to a smaller proportion of capital expenditure and a larger portion of current expenditure.

**Table:** Summary of forecast assumptions

| Annual percentage change | 2018* | 2019 | 2020 |
|---|---|---|---|
| Dubai crude oil price (U.S. dollar per barrel) | 69.6 | 64.0 (65.9) | 63.0 (65.0) |
| Farm income (% YoY) | 1.3 | 2.0 (2.0) | -1.0 (-0.4) |
| Government consumption at current price (billion baht)[1/] | 2,637 | 2,762 (2,763) | 2,929 (2,909) |
| Public investment at current price (billion baht)[1/] | 962 | 998 (1,011) | 1,080 (1,103) |
| Fed funds rate (% at year end) | 2.375 | 1.625 (2.125) | 1.375 (1.875) |
| Trading partners' GDP growth (% YoY)[2/] | 3.6 | 2.8 (3.0) | 2.7 (2.8) |
| Regional currencies (excl. China) vis-à-vis the U.S. dollar (index)[3/] | 153.8 | 157.3 (156.8) | 158.0 (156.9) |

Notes: [1/] Assumption includes spending on infrastructure investment plans
[2/] Weighted by each trading partner's share in Thailand's total exports
[3/] Increasing index represents depreciation, decreasing index represents appreciation
* Outturns
( ) Monetary Policy Report June 2019

## The value of merchandise exports was projected to contract in line with slowdown in trading partner economies and global trade volume.

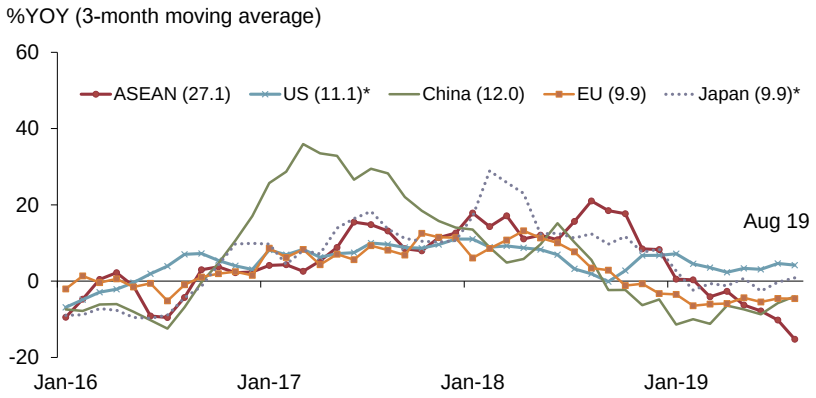
**The value of Thai merchandise exports in 2019 was expected to contract** mainly due to the decline in export volume following intensifying and prolonged trade tensions, which led to (1) slowdown in global trade volume in line with economic activities, as well as weaker consumption and investment confidence of the private sector in many countries, and (2) slow recovery of the electronics cycle. Meanwhile, export prices were also expected to decline in line with global crude oil prices. Moreover, the value of Thai merchandise exports (excluding gold) to most markets were expected to contract, except for the U.S. where Thai products would substitute Chinese imports (Chart 2.18). In addition, intensified trade tensions during the third quarter led to gold price surge and accelerated volume of gold exports, partially offsetting the contraction of the overall export value.

Nevertheless, merchandise exports in the period ahead were expected to recover, albeit gradually, thanks to the relocation of production base to Thailand in previous periods (Chart 2.19). Exports of hard disk drives started to show signs of improvement in the third quarter. Moreover, the U.S. would import more categories of Thai exports to substitute

Chinese imports including refrigerators and apparels, compared to the previous periods when only exports in auto tyres increased. **The Committee projected merchandise exports value to contract by 1.0 percent in 2019 and expand by 1.7 percent in 2020, lower than the estimates in the previous *Monetary Policy Report*.**

**Chart 2.18** Merchandise exports contracted in almost all export destinations

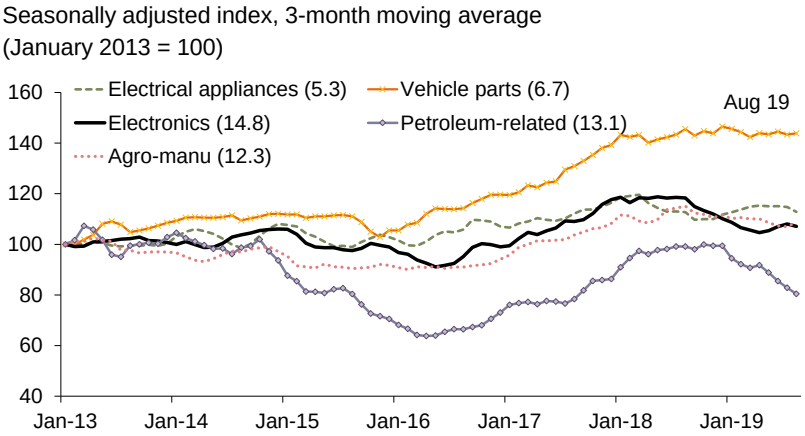
Value of Thai merchandise exports (excl. gold), by export destination




Note: Number in ( ) represents the share of total exports in 2018
* Excluding exports of munitions for Cobra Gold Military Exercise
Source: Customs Department, calculations by Bank of Thailand

**Chart 2.19** Thai merchandise exports continued to contract in various product categories, except exports of auto parts and electrical appliances which benefited from redirected orders and relocation of production base

Value of merchandise exports, by product category



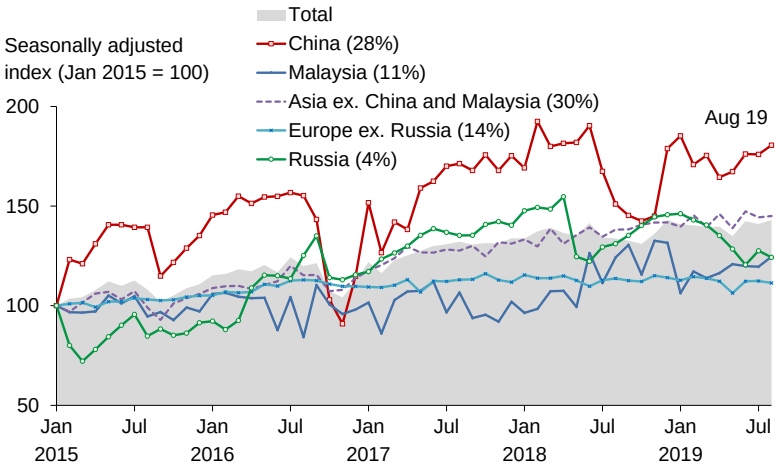

Note: Number in () denotes share of total exports in 2018
Source: Customs Department, calculations by Bank of Thailand

Exports of services were projected to decelerate in line with the global economic outlook.

**Exports of services were projected to decelerate** as tourism receipts stalled due to (1) lower foreign tourist figures across most nationalities (Chart 2.20) and (2) lower spending per person amid global economic slowdown. However, government measures to boost tourism through extending the exemption of visa-on-arrival (VOA) fees from the end of October 2019 to April 2020 would attract some more foreign tourists. **The projected foreign tourist figures were revised down to 39.7 in 2019 and 41.2 million in 2020, slightly lower than the estimates in the previous *Monetary Policy Report*.**

**Chart 2.20** Foreign tourist figures were expected to decline in many regions

Tourist figures, by nationalities




Note: Number in () denotes share of total tourists in 2018
Source: Ministry of Tourism and Sports, calculations by Bank of Thailand

**The current account was projected to record 34.2 and 30.4 billion U.S. dollar surplus in 2019 and 2020 respectively, higher than the projection in the previous *Monetary Policy Report*** due to (1) increased gold exports and (2) import slowdown in line with outlook of private investment and the imports of raw materials for export-oriented production.

Private consumption was expected to be weighed down by weakened purchasing power, despite a transient support by government stimulus measures.

**Private consumption in 2019 was projected to expand as previously estimated**, driven by additional government stimulus measures in the latter half of the year. These included measures to boost domestic tourism, to support low-income earners through higher cash limit of the social welfare card scheme, and to support farm income. However, factors supporting overall purchasing power weakened, including both non-farm income and consumer confidence. **In 2020, private consumption was projected to decelerate more than previously estimated.** Purchasing power would remain weak because employment and working hours especially in export-related manufacturing sectors would likely face greater impacts from export contraction, potentially weighing down on non-farm income further. Farm income next year was projected to contract from the high base effect of agricultural prices due to natural disasters this year. In addition, elevated household debt may exert additional pressures on private consumption, especially when household income faded. This may worsen

debt serviceability of households and lead to tightening credit standards of financial institutions, consequently weighing on private consumption in the period ahead.

Private investment would expand at lower rate than previously assessed, but remained on a positive outlook for next year.

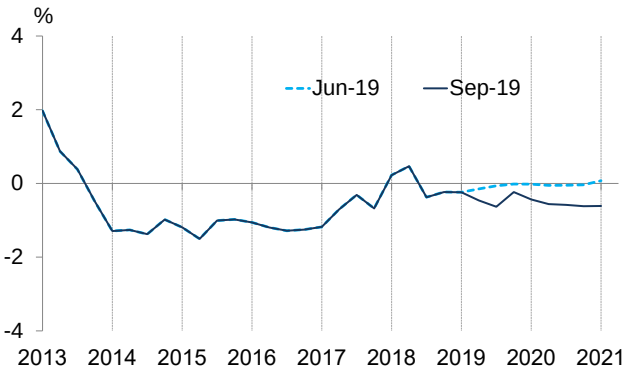
**Private investment would expand at lower rate than previously assessed** as export contraction and domestic demand slowdown began to exert impact on investment more apparently, as reflected by lower profitability and investment confidence of businesses. However, government measures to boost investment, including low lending rates for SMEs and the Thailand Plus Package, would help shore up investment to some extent. Looking forward, private investment was expected to benefit from additional relocation of production base to Thailand as businesses seek to avoid the impact of intensifying and prolonged trade tensions. Moreover, investor confidence was expected to pick up due to continued infrastructure investment in the Eastern Economic Corridor (EEC) as well as the public-private partnership (PPP) projects, as seen through continued investment promotion applications to the Board of Investment (BOI) and land sales for industrial purposes in the EEC.

Public spending would support economic growth to a lesser extent than previously assessed.

**Public spending was expected to expand less than expected** due to delays in state-owned enterprise investment projects, including the high-speed rail project between Bangkok and Nakhon Ratchasima which experienced a delay in the construction bid. Moreover, there was a change in budget structure of the Annual Budget Expenditure Act B.E. 2563 (A.D. 2020) that led to a smaller proportion of capital structure and a larger proportion of current expenditure. Public investment would subsequently decelerate, while government consumption would increase in 2020.

Projections of headline and core inflation were revised down in line with the negative output gap throughout the forecast horizon.

**Chart 2.21** Output gap

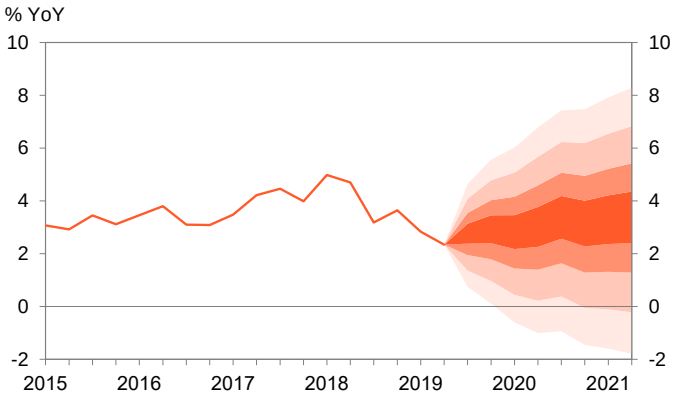



**Headline inflation projection for 2019 was expected to be below the inflation target and lower than the assessment in the previous *Monetary Policy Report*** due to greater-than-expected decline in energy prices between June and August 2019. Energy prices would decrease in line with global crude oil prices. Core inflation would soften in line with lower demand-pull inflationary pressures, reflected by the lower-than-expected and negative output gap throughout the forecast horizon (Chart 2.21). However, fresh food prices, such as fruit and glutinous rice, were expected to rise in the short term due to limited market supply, and would support headline inflation to some extent. Moreover, core inflation was projected to increase in the period ahead in line with the economic outlook and the increased cost-push inflationary pressures in 2020 from the prospects of minimum wage increase. **Thus, the Committee projected headline inflation to average 0.8 and 1.0 percent in 2019 and 2020, and core inflation to average 0.6 and 0.9 percent in 2019 and 2020, respectively.**

## Growth and inflation projections were subject to greater downside risks.

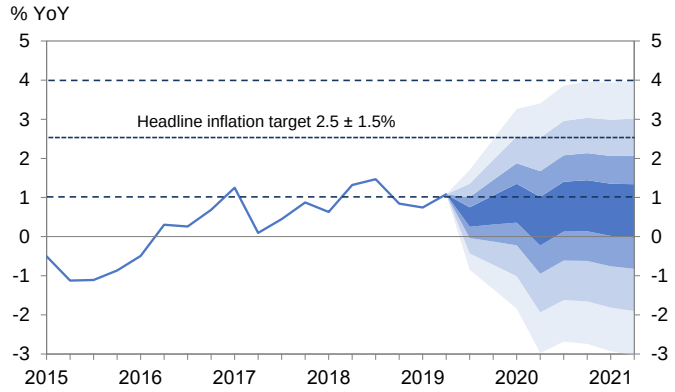
Under the Committee's assessment, **growth and inflation projections were subject to greater downside risks than the assessment in the previous *Monetary Policy Report***, as shown by the fan chart that tilted more downward (Chart 2.22), mainly due to external risks. **The possibilities that the Thai economy would underperform the baseline projection** would be due to (1) slower-than-expected trading partner economies and global trade volume as a result of intensifying trade tensions and impinging on Thai merchandise exports more than expected, (2) higher-than-expected increase in crude oil prices due to escalating tensions in the Middle East, (3) potential implementation delay in government policy and lack of continuation from the previous government, and (4) larger-than-expected household income shocks impinging on private consumption as a result of multiple factors including more-than-expected impact of exports on employment as well as the impact of intensifying natural disasters on farm income. On the other hand, **there were possibilities that the Thai economy would outperform the baseline projection attributable to** (1) higher-than-expected growth of trading partner economies if there were additional government stimulus measures, improvements in trade tensions, or faster-than-expected recovery in the electronics cycle, and (2) higher-than-expected domestic demand as a result of public infrastructure investment and PPP projects, which may boost private investment, as well as the potential implementation of additional government stimulus measures such as investment promotion policy. **Meanwhile, downside risks to headline and core inflation projections remained** (Chart 2.23 and 2.24), in line with risks to the growth projection.

**Chart 2.22** Growth forecast



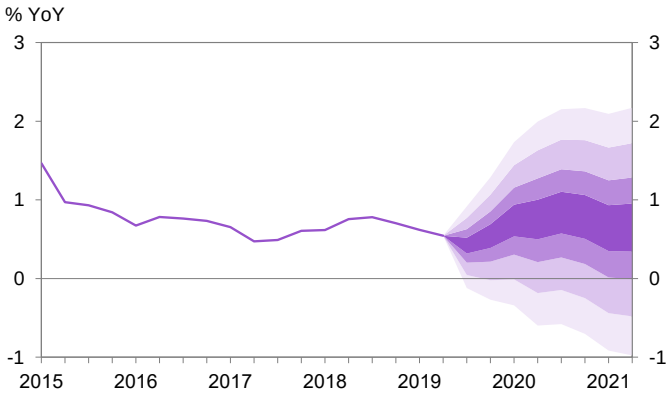

Note: Fan chart covers 90% of the probability distribution

**Chart 2.23** Headline inflation forecast




Note: Fan chart covers 90% of the probability distribution

**Chart 2.24** Core inflation forecast




Note: Fan chart covers 90% of the probability distribution

## Table 2.3 Forecasts of GDP and components

| Annual percentage change | 2018* | 2019 | 2020 |
|---|---|---|---|
| GDP growth | 4.1 | 2.8 (3.3) | 3.3 (3.7) |
| Domestic demand | 3.9 | 3.3 (3.5) | 3.7 (3.9) |
| Private consumption | 4.6 | 3.8 (3.8) | 3.1 (3.4) |
| Private investment | 3.9 | 3.0 (3.8) | 4.8 (5.5) |
| Government consumption | 1.8 | 2.3 (2.2) | 3.4 (2.6) |
| Public investment | 3.3 | 2.5 (3.8) | 6.3 (7.2) |
| Exports of goods and services | 4.2 | -2.3 (0.3) | 2.3 (3.9) |
| Imports of goods and services | 8.6 | -2.6 (0.7) | 2.9 (4.5) |
| Current account (billion, U.S. dollars) | 28.5 | 34.2 (29.1) | 30.4 (26.3) |
| Value of merchandise exports | 7.5 | -1.0 (0.0) | 1.7 (4.3) |
| Value of merchandise imports | 13.7 | -3.6. (-0.3) | 3.5 (4.8) |
| Number of foreign tourists (million person) | 38.2 | 39.7 (39.9) | 41.2 (41.3) |

Note: *Outturns

( ) Monetary Policy Report June 2019

# 3. Monetary Policy Decision

Accommodative monetary policy stance should support the rise of headline inflation toward target and contribute to the continuation of economic growth amid heightened external uncertainties. With regard to addressing financial stability risks, given softening growth outlook and prolonged low interest rates in the period ahead, microprudential and macroprudential measures should be appropriately combined and targeted.

**At the meeting on 7 August 2019, the Committee voted 5 to 2 to cut the policy rate by 0.25 percentage point from 1.75 to 1.50 percent. Two members voted to maintain the policy rate at 1.75 percent.** The Committee assessed that the Thai economy would expand at a lower rate than previously assessed and below potential due to a contraction in exports, which started to affect domestic demand. Headline inflation was projected to be lower than the lower bound of the inflation target. Financial stability risks had already been addressed to some extent, although there remained pockets of risks that warranted monitoring. Thus, most members viewed that a more accommodative monetary policy stance would contribute to the continuation of economic growth and should support the rise of headline inflation toward target. Nevertheless, two members viewed that the policy rate cut under the already accommodative monetary policy might not lend additional support to economic growth, compared with potentially increased financial stability risks. Moreover, there remained a need to preserve policy space. Moreover, the Committee would closely monitor developments of exchange rates and capital flows, as well as assess the necessity of additional appropriate measures.

**At the meeting on 25 September 2019, the Committee voted unanimously to maintain the policy rate at 1.50 percent.** The Committee assessed that projections of the Thai economy and inflation, as well as financial stability were in line with Committee's previous assessment in August. The Committee viewed that accommodative monetary policy stance would contribute to the continuation of economic growth and should support the rise of headline inflation toward target. Also, the Committee would closely monitor developments of exchange rates and capital flows, and would consider implementing additional appropriate measures if necessary. In addition, the Committee viewed that microprudential and macroprudential measures should be appropriately combined and targeted to ensure financial stability.

**Looking ahead**, the Committee would be data-dependent by closely monitoring the developments of the outlook for growth, inflation, and financial stability, together with associated risks, especially the impacts from trade tensions, in deliberating appropriate monetary policy going forward. The Committee would stand ready to use policy tools as appropriate. Nevertheless, the Thai economy continued to face structural problems, which would affect competitiveness and growth outlook going forward. This should be firmly addressed by all related parties.

In determining the most appropriate course of monetary policy, the Committee placed great emphasis on economic growth, development of headline inflation, and preserving financial stability. Details on the key assessment are as follows.

## The Thai economy and inflation were expected to expand at a lower rate than previously assessed.

Developments of economic indicators in the third quarter of 2019 reflected that economic activities related to merchandise exports, manufacturing, and investment would slow down more than previously assessed. The spillovers began to be observed on private consumption and employment in export-related manufacturing sectors. Furthermore, the output gap was expected to be more negative. That is, actual output would remain below potential throughout the forecast horizon, reflecting a decline in demand-pull inflationary pressures. This led in part to a downward revision of headline inflation forecast in 2019, which would be below the lower bound of the inflation target. Looking ahead, the Thai economy would still face downside risks from trade tensions and geopolitical risks, which were highly likely to be prolonged and intensify, causing adverse impacts on the global economy and trade volume. Meanwhile, public expenditure would not lend full support to economic growth due to delayed enactment of the Annual Budget Expenditure Act B.E. 2563 (A.D. 2020), as well as the postponement of some state-owned enterprise investment projects, despite additional government stimulus measures in the second half of this year. Against this backdrop, **the Committee viewed that a more accommodative monetary policy stance should contribute to the continuation of economic growth and support the rise of headline inflation toward target, allowing both public and private sectors time to adjust themselves to cope with periods of slower economic growth.**

## The Committee was concerned and closely monitored impact of the baht on the economy.

Over recent periods, the baht continued to strengthen mainly on account of a large current account surplus and foreign direct investment (FDI) inflows. However, there were net capital outflows from the bond and stock markets in the recent period, partly as a result of the Bank of Thailand's measures to enhance surveillances of short-term capital flows in order to limit channels for speculation on the baht appreciation. The Committee assessed that a continued appreciation of the baht could have a more pronounced impact on exports and the overall economy in the period of slower trading partners' growth and stronger international competition. Signs of such impact started to be seen on corporate profitability and liquidity, debt serviceability, and a reduction in overtime employment in export-related manufacturing sectors. **The Committee remained concerned about the baht appreciation against trading partner currencies and saw a need to closely monitor impacts from developments of exchange rates and capital flows. The Committee would consider implementing additional measures at an appropriate timing if necessary.** The additional measures included (1) those under the Bank of Thailand's authority, such as continued relaxation of capital outflow regulations to encourage Thai residents to increase their portfolio investment abroad, and (2) those required collaboration with other organizations, including efforts to stimulate investment to reduce elevated current account surplus.

## Some financial stability risks abated, but there remained pockets of risks that warranted monitoring and should be addressed with various targeted measures.

The Committee assessed that financial stability remained sound overall. However, there remained a need to monitor risks that could pose vulnerabilities to financial stability in the future. An assessment of various types of financial stability indicators such as real estate price indices, components of household debt, stress testing, scenario planning, and financial cycle (Box: Financial cycle and its policy implications) revealed that some financial stability risks abated,

following the implementation of the LTV measure in an earlier period and softening economic growth outlook. This was reflected by a slower growth in mortgage loans in the second quarter of 2019, more tightening lending standards among financial institutions, postponement of new project launches by some real estate developers amid slower economic growth, and declining residential prices. Nevertheless, some risks had not improved. These included, risks from vulnerabilities to income shocks during periods of economic slowdown, such as risks arising from debt serviceability of households and SMEs with weak financial position, and risks in the real estate sector due to oversupply in certain areas, with additional impact on developers due to domestic and external demand slowdown. Second, the search-for-yield behavior in the prolonged low interest rate environment, especially related to saving cooperatives and leverages of large corporates, could lead to underpricing of risks. This also included potential changes in household saving behavior and debt accumulation given low interest rate environment.

**The Committee expressed concerns about elevated household debt that was expected to rise further, as it could lead to a build-up of vulnerabilities among households.** Thus, the Committee would closely monitor the debt service ratio (DSR) data in accordance with the standardized calculation methodology agreed between the Bank of Thailand and commercial banks, where financial institutions would start reporting the data in the forth quarter of 2019. In addition, the Committee would monitor debt serviceability of households and SMEs, which was expected to deteriorate during economic slowdown and could affect their income. Nevertheless, the Committee viewed that addressing household debt would require understanding of different groups of households, such as households with strong financial positions, households with high debt and vulnerability, and households without financial discipline. Such understanding would be crucial in considering targeted and appropriate approaches for different groups of households. The Committee also deemed it necessary for the Bank of Thailand to cooperate with relevant agencies to push forward approaches to address household and SMEs debt problems, such as debt restructuring and debt clinic, promoting appropriate application of responsible lending practices, as well as cultivating financial discipline and saving behavior among households.

# Financial cycle and its policy implications

Since the Global Financial Crisis (GFC) in 2008, most central banks have put more emphasis on preserving financial stability (FS), perceiving that the economy cannot grow in a sustainable way, disregarding a stable financial system. A systematic assessment of financial stability risks within monetary policy deliberation processes, and the availability of indicators that can accurately signal those risks, have consequently become of great importance. **This article aims to explain the construction of a financial cycle indicator, which will serve as a tool to measure Thailand's overall FS risks together with other existing FS indicators and hence become part of appropriate monetary policy consideration.**

## 1. The concept of a financial cycle indicator

**Risks to financial stability can be assessed through a financial cycle (FC) indicator, which measures deviations of the level of credit and asset prices from their respective trends. FC, thus, shows phases of expansion and contraction of the financial sector in each period.** A positive value of FC indicates the acceleration of credit and asset price growth that may be inconsistent with the economy's growth potential. On the other hand, a negative value of FC reflects the expansion of credit and asset prices that is below its trend, which normally occurs after a financial crisis.

**In fact, dynamics of credit and asset prices are reinforcing.** During periods of eased monetary conditions, low interest rates will boost credit growth. Given that the credit expansion in part attributes to investment in money, capital, and real estate markets, it could lead to higher asset prices. The rise in asset prices then helps increase the collateral value, thereby contributing to further credit expansion.

## 2. The financial cycle indicator for Thailand

**The FC indicator reflects deviations of financial variables[14/] from their trends. For Thailand, the financial cycle is a composite index, constructed by averaging the cycles of 8 financial indicators.[15/]** These include 4 credit variables and 4 asset-price variables.[16/] Key observations are as follows:

**(1) The peak of FC portends the occurrence of a financial crisis.** Prior to the Asian Financial Crisis in 1997, the FC was positive and stood at a high level (Chart 2). Nevertheless, in gauging the extent of risks during financial crisis, the amplitude of a cycle should also be taken into consideration. For example, the current level of Thailand's FC already reached its peak in 2016, the amplitude of which was lower than that before 1997. To assess the concerned level of FC, one

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[14/] FC is calculated by a one-sided band-pass filter, namely Christiano-Fitzgerald (CF) filter. The selection of frequencies corresponds to the financial-cycle length of around 8 to 30 years, which is suitable for measuring risks of financial crisis that usually take a while to accumulate and infrequently occur.

[15/] The criteria for choosing appropriate financial variables include: (1) such variables can reflect leverage and asset pricing in the financial system (2) their data cover periods of Thailand's past financial crisis and (3) cycles of the chosen variables are overall consistent with one another.

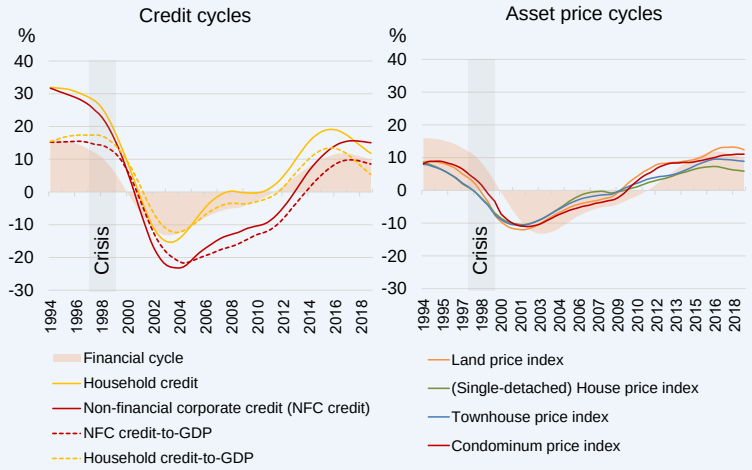
[16/] Credit indicators, which can reflect leverage in the financial system, comprise (1) outstanding credit extended to non-financial corporations, (2) the ratio of total credit extended to non-financial corporations to GDP, (3) outstanding credit extended to households, and (4) the ratio of total household credit to GDP. Meanwhile, asset-price indicators, which serve to signal underpricing of risks, include (1) the single-detached house (including land) price index, (2) townhouse (including land) price index, (3) land price index, and (4) condominium price index. However, equity prices, i.e. the SET index, are not included in the construction of Thailand's FC, since their cycles have exhibited relatively high volatility compared to the 8 variables aforementioned.

must consider linkages between FC and the probability of crisis, which will be discussed in the next section.

**(2) Assessing the components of FC, whose dynamics may differ in some periods, helps enhance our understanding about the contributing factors to systemic risks.** For example, (1) cycles of household credit (as a proportion of GDP) accelerated during 2011. This reflects the level of Thailand's household debt that heightened after the great flood, as well as the launch of government measures to stimulate consumption, such as first-car and first-home buyer schemes, and a rice-pledging scheme, which induced financial institutions to extend more loans (Chart 1, left-hand panel). (2) Land and condominium price cycles increased markedly during 2009-2010, reflecting greater demand for condominiums than other forms of real estate. This prompted the Bank of Thailand to impose regulations on mortgage lending (LTV) in 2011 (Chart 1, right-hand panel).

**Chart 1** Credit cycles and asset price cycles are coherent

Deviation from trend




Source: Calculations by Bank of Thailand

## 3. The relationship between FC and the probability of financial crisis

**The level of FC signals the probability of crisis, which could serve as an early warning indicator within the assessments of FS risks.** This article applies the ratio of credit outstanding to GDP and real estate prices from both advanced and emerging economies to estimate such probability of crisis[17/] in future periods. The measure will allow policymakers sufficient time to implement macroprudential measures should FS risks materialize.

**The estimation results suggest that the probability of crisis for Thailand, the U.S. and the United Kingdom synchronizes well with FC. The probability of crisis, in most cases, also surges to a very high level ahead of a financial crisis,** whereas it appears low whenever FC stands at a low level or prevails within a negative territory. Furthermore, it can be observed that the probabilities of crisis for Thailand and the U.S. prior to a financial crisis are relatively similar, although their FC level markedly differs.

17/ The estimation is based upon a panel logistic regression model to assess the probability of crisis one to three years ahead. The samples are from both advanced economies (AEs), including the U.S., the United Kingdom, France, Italy, Netherlands, and Spain, and emerging market economies (EMs), i.e. Thailand and Malaysia. The identification of periods during which financial crises took place in each country is based on Laeven and Valencia (2013) "Systemic Banking Crises Database: An Update". The cross-country regression should help improve accuracy of the estimates, since a financial crisis in Thailand occurred just once since 1993 and the estimation using only Thai data yields a statistically insignificant estimate.

## 4. Policy implications

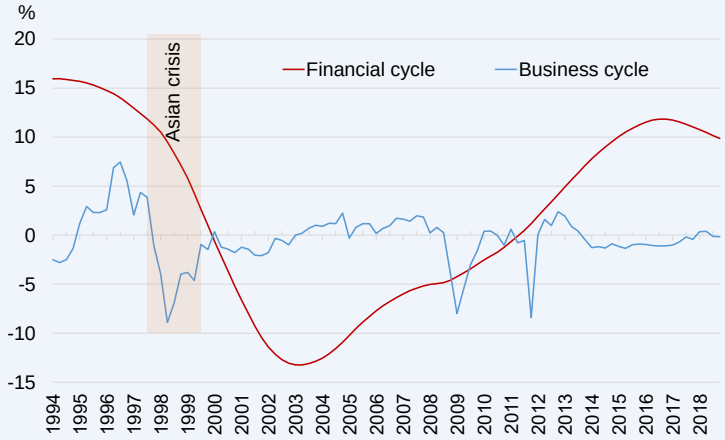
**Evaluating FC in conjunction with business cycles (output gap)[18/] enables more comprehensive assessments of risks to both economic growth and financial stability in monetary policy conduct.** This is because either an increase or decrease in the policy rate simultaneously impacts the economy and financial stability. For instance, in periods when FC expands considerably or reaches a high level during a business cycle contraction, monetary policy easing aiming at stimulating the economy may lead to further accumulation of financial stability risks because lower borrowing costs encourage a more credit expansion and rising asset prices.

**A comparison of Thailand's FC and business cycles (Chart 2) suggests that (1) the duration and amplitude of FC are higher than those of business cycles.** A financial crisis normally stems from the accumulation of risks for extended periods. Once a crisis occurs, the economy faces a severe impact and takes time to recover. Meanwhile, an economic slowdown during a business-cycle contraction will take a shorter recovery period, since the economy is usually supported by demand-stimulating measures. **(2) Recessions that occur at the same time as a financial crisis tend to be more severe and protracted than ones without it,** since most financial institutions restrict their loans extension, making it difficult to boost demand and support economic recovery.

**The longer duration of FC than that of business cycles requires the conduct of monetary policy to be more forward-looking** in order to strike the right balance between maintaining macroeconomic stability and financial stability, given the differences in their cycle durations.

**Chart 2** Comparison of financial cycle and business cycle in Thailand

Deviation from trend




Note: Business cycle is expressed by output gap

Source: Calculations by Bank of Thailand

To conclude, the assessments of a financial cycle indicator and the probability of crisis together with other financial stability indicators, such as property price index, components of household debt, stress test and scenario planning, will help enhance the efficiency of monetary policy conduct and its capability to preserve FS in a more systematic manner.

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[18/] Business cycles, measured as deviations of GDP from its trend, show developments of economic conditions during both expansion and contraction phases, which evolve in tandem with economic activities in each period. Whereas business cycles measure the level of an economic variable relative to its short-term trend, financial cycles show the level of financial variables relative to their corresponding medium-term trend.

# The concept of growth at risk and the linkage to Thailand's financial stability

**Growth at Risk (GaR) is a tool used to measure risks to economic growth in the future.** It applies the concept of Value at Risk, an estimate of financial risks, to the evaluation of downside risks to economic growth. **The objective of this article is to explain the measurement of GaR for Thailand as well as the empirical relationship between GaR and a financial cycle (FC), both of which are crucial elements for the conduct of monetary policy that systematically incorporates financial stability (FS) into considerations.**

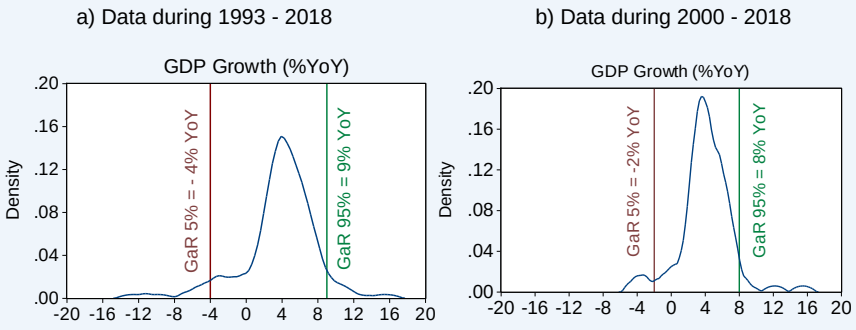
## 1. Measuring GaR

The forward-looking assessment of the economy necessarily take into consideration both downside and upside risks that may deviate economic growth from baseline projection. GaR is one particular tool that can measure such risks by forecasting one-year-ahead quarterly GDP growth at different percentiles, where **GDP growth at X percentile (GaR X%) implies the probability of X percent that the GDP growth will fall below GaR X%**[19/]. In general, GaR 5% is chosen to reflect extreme downside risks or crisis events which infrequently occur[20/]. Moreover, GaR can be useful in assessing upside risks through the estimate of GaR 95%.

Based on the historical distribution of Thailand's quarterly GDP growth during 1993-2018 (Chart 1a), GaR 5% registers at -4 percent (red vertical line). That is, there is a 5-percent chance that the Thai economy going forward will contract by more than 4 percent. Past observations featuring GDP growth below GaR 5% were mostly in the periods of 1997 Asian Financial Crisis (AFC), whereas GaR 95% stands at around 9 percent. However, since 2000 (Chart 1b), it can be observed that the historical distribution of GDP growth has narrowed. While GaR 95% remains close to the previous level at 8 percent, GaR 5% is at -2 percent. The episodes where the Thai economy contracted by over 2 percent correspond to the periods when Thailand was hit by the Global Financial Crisis (GFC) during 2008-2009 and the Great Flood in 2011.

**Chart 1** GaR 5%, GaR 95%, and distribution of Thailand's GDP growth

Distribution of Thailand's GDP growth in the past




Source: Office of the National Economic and Social Development Council (NESDB), calculations by Bank of Thailand

19/ Mathematically, GaR X% is the threshold that satisfies $Prob(y_{t+h} \leq GaR_{t+h}\ X\%) = X\%$, where $y_{t+h}$ denotes GDP growth h-quarter ahead. Normally, it is assumed that h=4, which amounts to measuring economic growth in the same quarter one-year ahead.

20/ Extreme events that drag economic growth below GaR 5% have a 5-percent chance of occurrence. In other words, they take place on average every 20 year.

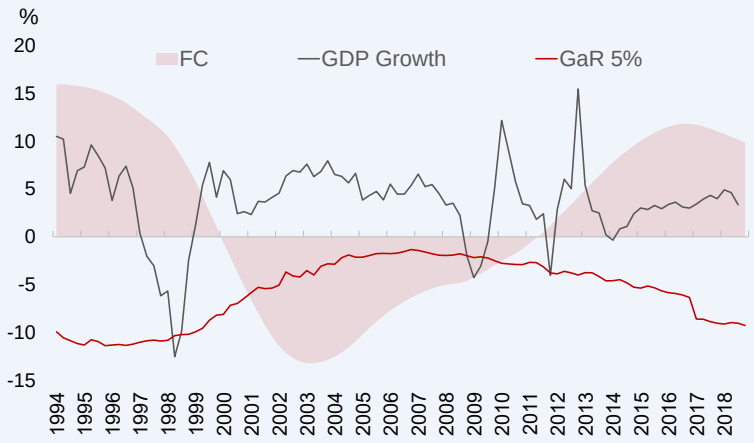
The above examples suggest that, **although the estimate of GaR 5% may alter by periods of assessment, the observations that are below GaR 5% well reflect severe recession events. Such events include financial crises, which are systemic risks to FS** that do not frequently occur but can have large negative impact on the economy once they occur. Given the fact that the probability of systemic risks to FS can be signaled by the FC indicator[21/], this naturally leads to the hypothesis that GaR is negatively correlated with FC. That is, an increase (decrease) in FC reflects heightened (lowered) systemic risks, prompting GaR 5% in future periods to deteriorate (improve).

## 2. The relationship between GaR and FC

**The empirical relation between GaR and FC can be studied through a quantile regression analysis,** where the current level of FC is used to predict GaR 5% in the future[22/], consistent with the forward-looking risk assessment. However, the estimation of Thailand's GaR absolutely depends on FC, i.e. excluding macroeconomic and external factors, and hence can only reflect risks emanated from the financial system.

The empirical findings suggest that **GaR 5% for Thailand, an indicator of downside risks, is negatively and significantly correlated with FC,** consistent with the hypothesis. Periods under which GDP growth outturn is lower than the predicted value of GaR 5% coincide with AFC, GFC and the great flood of 2011 (Chart 2). Meanwhile, GaR 95%, which portrays upside risks, is relatively constant and does not correlate with FC. It can, hence, be concluded that **systemic risks to FS have a connection with downside risks to future economic growth.** Therefore, mitigating FS risks is crucial in reducing downside risks to economic growth and the severity of financial crisis that could occur in the future, thereby providing more chances of achieving sustainable economic growth.

**Chart 2** Changes in GaR 5% as explained by Thailand's financial cycle




Source: Office of the National Economic and Social Development Council (NESDB), calculations by Bank of Thailand

21/ Additional details can be found in the box "Financial cycle and its policy implications (*Monetary Policy Report,* the third quarter of 2019)

22/ The baseline equation used in the study is $GaR_{t+4}X\% = \beta_0^{X\%} + \beta_1^{X\%} FC_t + \varepsilon_{t+4}$. The current value of FC is to predict GaR 5% four-quarter ahead. Other factors can also be taken into considerations in this equation, including the upcycle and downcycle of FC, and the non-linear relationship between FC and GaR. It is estimated by a cross-country panel-quantile regression, using the dataset of 9 countries. The estimation for only Thai data finds a negative $\beta_1$ coefficient.

## 3. Policy implications

The linkage between the GaR framework and the level of FC, which is an indicator of systemic risks to FS, is an important element for the systematic assessment of FS risks in monetary policy decision processes. In practice, the assessment of economic outlook should be performed in parallel with evaluating downside GAR, in order to enhance the comprehensiveness of the information set in support of the deliberation processes. **Nevertheless, the conduct of monetary policy could face intertemporal trade-offs in some occasions.** It would be beneficial to set the policy rate at a level that is not overly-expansionary, preventing the FC from rising too high and inducing risks to FS (the action referred to as “lean against the wind”). In turn, such level of policy rate would help reduce downside risks to future economic growth. Such policy decision, nevertheless, could affect economic growth in the short-term. On the contrary, monetary policy expansion, while being conducive to short-term economic growth, could have adverse consequences on economic and financial stability over the longer term. **The appropriate conduct of monetary policy should, therefore, aim at striking the balance between both short-term and long-term economic stability.**

# 4. Appendix

## 4.1 Table

### Thai Economy Dashboard

| Percent | 2017 | 2018 | 2018 | | | | 2019 | |
|---|---|---|---|---|---|---|---|---|
| | | | Q1 | Q2 | Q3 | Q4 | Q1 | Q2 |
| GDP growth | 4.0 | 4.1 | 5.0 | 4.7 | 3.2 | 3.6 | 2.8 | 2.3 |
| Production | | | | | | | | |
| Agriculture | 3.7 | 5.1 | 6.8 | 10.5 | 3.2 | 0.7 | 1.7 | -1.1 |
| Non-agriculture | 4.1 | 4.0 | 4.8 | 4.2 | 3.1 | 4.0 | 2.9 | 2.6 |
| Manufacturing | 3.0 | 3.0 | 3.7 | 3.1 | 1.5 | 3.5 | 0.6 | -0.2 |
| Construction | -2.8 | 2.7 | 1.2 | 1.9 | 4.5 | 3.4 | 3.0 | 3.4 |
| Wholesales and retail trade | 7.0 | 7.3 | 7.0 | 7.4 | 7.5 | 7.5 | 6.8 | 5.9 |
| Accommodation and Food Service | 10.6 | 7.9 | 13.1 | 8.8 | 4.1 | 5.3 | 4.9 | 3.7 |
| Transport and storage | 8.2 | 5.7 | 7.1 | 5.4 | 4.5 | 5.4 | 3.5 | 2.5 |
| Information and Communication | 2.7 | 6.8 | 5.2 | 7.7 | 7.4 | 6.9 | 6.5 | 9.3 |
| Financial intermediation | 5.4 | 3.3 | 3.5 | 4.6 | 3.1 | 1.8 | 2.0 | 1.8 |
| Real estate, renting, and business activities | 6.5 | 5.2 | 6.0 | 4.0 | 5.9 | 4.9 | 4.7 | 3.1 |
| Expenditure | | | | | | | | |
| Domestic demand | 2.1 | 3.9 | 3.3 | 3.6 | 4.3 | 4.3 | 4.1 | 3.2 |
| Private consumption | 3.0 | 4.6 | 3.8 | 4.1 | 5.2 | 5.4 | 4.9 | 4.4 |
| Private investment | 2.9 | 3.9 | 3.1 | 3.1 | 3.8 | 5.5 | 4.4 | 2.2 |
| Government consumption | 0.1 | 1.8 | 1.8 | 2.3 | 1.9 | 1.4 | 3.4 | 1.1 |
| Public investment | -1.2 | 3.3 | 4.0 | 4.9 | 4.2 | -0.1 | -0.1 | 1.4 |
| Imports of goods and services | 6.2 | 8.6 | 9.1 | 8.8 | 11.0 | 5.7 | -0.1 | -2.7 |
| imports of goods | 7.4 | 8.1 | 10.4 | 7.9 | 9.9 | 4.5 | -2.6 | -3.4 |
| imports of services | 1.3 | 10.7 | 3.9 | 12.8 | 16.1 | 10.4 | 10.6 | 0.1 |
| Exports of goods and services | 5.4 | 4.2 | 8.0 | 9.6 | -0.9 | 0.7 | -6.1 | -6.1 |
| exports of goods | 5.7 | 4.1 | 7.2 | 9.5 | -0.5 | 0.8 | -5.9 | -5.8 |
| exports of services | 4.6 | 4.4 | 9.9 | 10.3 | -2.2 | 0.0 | -7.2 | -7.0 |
| Trade balance (billion, U.S. dollars) | 32.6 | 22.4 | 7.4 | 6.4 | 3.9 | 4.6 | 6.5 | 5.7 |
| Current account (billion, U.S. dollars) | 44.0 | 28.5 | 14.2 | 4.2 | 3.8 | 6.3 | 12.7 | 6.7 |
| Financial account (billion, U.S. dollars) | -12.5 | -14.9 | -3.3 | -8.1 | -1.1 | -2.5 | -5.1 | -3.2 |
| International reserves (billion, U.S. dollars) | 202.6 | 205.6 | 215.6 | 206.8 | 204.5 | 205.6 | 212.2 | 215.8 |
| Unemployment rate (%) | 1.2 | 1.1 | 1.2 | 1.1 | 1.0 | 0.9 | 0.9 | 1.0 |
| Unemployment rate, seasonally-adjusted (%) | n.a. | n.a. | 1.2 | 1.0 | 1.0 | 1.0 | 0.9 | 0.9 |

Source: Office of the National Economic and Social Development Board National Statistical Office and Bank of Thailand

## Financial Stability Dashboard

| Indicators | 2017 | 2018 | 2018 | | | | 2019 | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | Q1 | Q2 | Q3 | Q4 | Q1 | Q2 | Jul | Aug |
| 1. Financial market sector | | | | | | | | | | |
| Bond market | | | | | | | | | | |
| Bond spread (10 years - 2 years) | 1.1 | 0.7 | 1.2 | 1.1 | 0.9 | 0.7 | 0.7 | 0.4 | 0.2 | 0.1 |
| Equity market | | | | | | | | | | |
| SET index (end of period) | 1,753.7 | 1,563.9 | 1,776.3 | 1,595.6 | 1,756.4 | 1,563.9 | 1,638.7 | 1,730.3 | 1,712.0 | 1,654.9 |
| Actual volatility of SET index[1/] | 6.5 | 12.1 | 9.4 | 12.3 | 11.6 | 14.0 | 8.4 | 7.8 | 8.3 | 14.8 |
| Price to Earnings ratio (P/E ratio) (times) | 19.1 | 14.8 | 18.3 | 16.2 | 17.3 | 14.8 | 17.1 | 18.6 | 18.5 | 18.8 |
| Exchange rate market | | | | | | | | | | |
| Actual volatility of Thai baht (%annualized)[2/] | 4.4 | 3.3 | 4.6 | 4.4 | 4.5 | 4.7 | 4.6 | 4.5 | 5.7 | 4.4 |
| Nominal Effective Exchange Rate (NEER) | 110.6 | 115.6 | 114.8 | 115.2 | 115.4 | 117.0 | 120.0 | 121.2 | 123.6 | 125.2 |
| Real Effective Exchange Rate (REER) | 103.7 | 107.2 | 106.3 | 107.1 | 107.2 | 108.2 | 110.2 | 111.6 | 113.6 | 114.7 |
| 2. Financial institution sector[3/] | | | | | | | | | | |
| Minimum Lending Rate (MLR)[4/] | 6.28 | 6.28 | 6.28 | 6.28 | 6.28 | 6.28 | 6.28 | 6.28 | 6.28 | 6.28 |
| 12-month fixed deposit rate[4/] | 1.37 | 1.37 | 1.37 | 1.37 | 1.37 | 1.37 | 1.42 | 1.46 | 1.46 | 1.46 |
| Capital adequacy | | | | | | | | | | |
| Capital funds / Risk-weighted asset (%) | 18.2 | 18.34 | 18.09 | 17.86 | 18.39 | 18.34 | 18.22 | 18.00 | 18.27 | n.a. |
| Earning and profitability | | | | | | | | | | |
| Net profit (billion, Thai baht) | 187.1 | 207.25 | 50.21 | 56.38 | 51.12 | 49.54 | 57.08 | 60.80 | n.a. | n.a. |
| Return on assets (ROA) (times) | 1.1 | 1.13 | 1.11 | 1.17 | 1.16 | 1.13 | 1.21 | 1.20 | n.a. | n.a. |
| Liquidity | | | | | | | | | | |
| Loan to Deposit and B/E (%) | 96.1 | 98.16 | 95.01 | 96.83 | 98.23 | 98.16 | 96.76 | 97.90 | 97.36 | n.a. |
| 3. Household sector | | | | | | | | | | |
| Household debt to GDP (%) | 78.4 | 78.6 | 77.8 | 77.7 | 77.9 | 78.6 | 78.7 | 78.7 | n.a. | n.a. |
| Financial assets to debt (times) | 2.6 | n.a. | 2.7 | 2.6 | 2.5 | 2.4 | 2.5 | 2.4 | n.a. | n.a. |
| Non-Performing Loans (NPLs) of commercial banks (%) | | | | | | | | | | |
| Consumer loans | 2.7 | 2.7 | 2.8 | 2.7 | 2.7 | 2.7 | 2.8 | 2.7 | n.a. | n.a. |
| Housing loans | 3.2 | 3.2 | 3.4 | 3.4 | 3.4 | 3.2 | 3.3 | 3.3 | n.a. | n.a. |
| Auto leasing | 1.6 | 1.7 | 1.5 | 1.5 | 1.6 | 1.7 | 1.7 | 1.8 | n.a. | n.a. |
| Credit cards | 2.6 | 2.3 | 3.2 | 2.4 | 2.5 | 2.3 | 2.5 | 2.2 | n.a. | n.a. |
| Other personal loans | 2.5 | 2.5 | 2.7 | 2.5 | 2.5 | 2.5 | 2.6 | 2.5 | n.a. | n.a. |
| 4. Non-financial corporate sector[5/] | | | | | | | | | | |
| Operating profit margin (OPM) (%) | 8.0 | 7.5 | 7.9 | 7.8 | 8.0 | 6.5 | 7.5 | 5.9 | n.a. | n.a. |
| Debt to Equity ratio (D/E ratio) (times) | 0.7 | 0.7 | 0.7 | 0.7 | 0.8 | 0.7 | 0.7 | 0.8 | n.a. | n.a. |
| Interest coverage ratio (ICR) (times) | 6.5 | 6.4 | 7.1 | 6.5 | 6.6 | 5.4 | 5.2 | 3.8 | n.a. | n.a. |
| Current ratio (times) | 1.7 | 1.6 | 1.7 | 1.7 | 1.6 | 1.6 | 1.6 | 1.6 | n.a. | n.a. |
| Non-Performing Loans (NPLs) of commercial banks (%) | | | | | | | | | | |
| Large businesses | 1.8 | 1.7 | 1.7 | 1.7 | 1.5 | 1.7 | 1.5 | 1.7 | n.a. | n.a. |
| SMEs | 4.4 | 4.5 | 4.5 | 4.5 | 4.7 | 4.5 | 4.6 | 4.5 | n.a. | n.a. |

Note:

[1/] Calculated by 'annualized standard deviation of return' method

[2/] Daily volatility (using exponentially weighted moving average method)

[3/] Based on data of all commercial banks

[4/] Average value of 5 largest Thai commercial banks

[5/] Only listed companies on Stock Exchange of Thailand (median value); with data revisions

## Financial Stability Dashboard (continue)

| Indicators | 2017 | 2018 | 2018 | | | | 2019 | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | Q1 | Q2 | Q3 | Q4 | Q1 | Q2 | Jul | Aug |
| 5. Real estate sector | | | | | | | | | | |
| Number of approved mortgages from commercial banks (Bangkok and Vicinity) (units) | | | | | | | | | | |
| Total | 65,124 | 74,924 | 15,012 | 18,352 | 19,194 | 22,366 | 20,260 | 16,056 | 5,223 | 5,465 |
| Single-detached and semi-detached houses | 14,517 | 16,019 | 3,527 | 3,858 | 3,990 | 4,644 | 4,484 | 3,467 | 1,217 | 1,267 |
| Townhouses and commercial buildings | 21,469 | 25,240 | 5,167 | 6,099 | 6,391 | 7,583 | 7,385 | 5,738 | 1,870 | 1,909 |
| Condominiums | 29,138 | 33,665 | 6,318 | 8,395 | 8,813 | 10,139 | 8,391 | 6,851 | 2,136 | 2,289 |
| Number of new housing units launched for sale (Bangkok and Vicinity) (units) | | | | | | | | | | |
| Total | 110,575 | 114,477 | 26,829 | 19,836 | 42,451 | 35,861 | 29,498 | 25,999 | 8,374 | 7,858 |
| Single-detached and semi-detached houses | 19,433 | 14,280 | 4,223 | 2,790 | 6,248 | 5,050 | 4,458 | 4,213 | 899 | 914 |
| Townhouses and commercial buildings | 32,792 | 36,571 | 6,657 | 6,548 | 8,759 | 10,385 | 5,236 | 7,940 | 1,980 | 3,534 |
| Condominiums | 58,350 | 63,626 | 15,949 | 10,498 | 27,444 | 20,426 | 19,804 | 13,846 | 5,495 | 3,410 |
| Housing price index (2009 = 100) | | | | | | | | | | |
| Single-detached houses (including land) | 130.9 | 138.8 | 137.8 | 138.0 | 139.5 | 139.0 | 141.9 | 142.9 | 143.3 | 144.3 |
| Townhouses (including land) | 141.1 | 149.9 | 146.0 | 149.6 | 151.2 | 151.1 | 156.6 | 154.7 | 153.2 | 153.0 |
| Condominiums | 169.6 | 180.9 | 179.8 | 177.6 | 180.1 | 185.9 | 181.6 | 177.0 | 178.7 | 178.1 |
| Land | 171.7 | 175.8 | 170.4 | 173.5 | 172.9 | 177.6 | 175.5 | 168.1 | 168.7 | 168.0 |
| 6. Fiscal sector | | | | | | | | | | |
| Public debt to GDP (%) | 40.7 | 41.2 | 41.2 | 41.0 | 42.1 | 41.9 | 41.9 | 41.4 | 41.5 | n.a. |
| 7. External sector | | | | | | | | | | |
| Current account balance to GDP (%) | 9.6 | 5.6 | 11.0 | 3.4 | 3.1 | 4.9 | 9.5 | 5.0 | n.a. | n.a. |
| External debt to GDP (%)[6/] | 36.7 | 35.5 | 36.5 | 35.1 | 35.5 | 35.5 | 35.1 | 34.8 | n.a. | n.a. |
| External debt (billion, U.S. dollars) | 155.2 | 162.4 | 157.5 | 154.2 | 159.0 | 162.4 | 164.5 | 166.8 | 166.7 | 168.8 |
| Short-term (%) | 44.3 | 39.0 | 42.9 | 42.9 | 41.6 | 39.0 | 38.5 | 37.4 | 36.9 | 37.0 |
| Long-term (%) | 55.7 | 61.0 | 57.1 | 57.1 | 58.4 | 61.0 | 61.5 | 62.6 | 63.1 | 63.0 |
| International reserves / Short-term external debt (times) | 2.9 | 3.2 | 3.2 | 3.1 | 3.1 | 3.2 | 3.4 | 3.5 | 3.5 | 3.5 |

Note:

[6/] External debt / 3-year average nominal GDP

Table: Probability distribution of GDP growth forecast

| Percent | 2562 | | 2563 | | | | 2564 | | |
|---|---|---|---|---|---|---|---|---|---|
| | Q2 | Q3 | Q4 | Q1 | Q2 | Q3 | Q4 | Q1 | Q2 |
| > 9 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 2 | 3 |
| 8-9 | 0 | 0 | 0 | 0 | 1 | 2 | 2 | 3 | 3 |
| 7-8 | 0 | 0 | 0 | 1 | 3 | 4 | 4 | 5 | 5 |
| 6-7 | 0 | 0 | 2 | 4 | 6 | 7 | 7 | 8 | 8 |
| 5-6 | 0 | 2 | 7 | 8 | 10 | 11 | 10 | 11 | 10 |
| 4-5 | 0 | 12 | 16 | 14 | 14 | 14 | 13 | 13 | 12 |
| 3-4 | 0 | 28 | 23 | 19 | 17 | 16 | 15 | 14 | 13 |
| 2-3 | 100 | 32 | 23 | 19 | 16 | 14 | 14 | 13 | 12 |
| 1-2 | 0 | 19 | 16 | 16 | 13 | 12 | 12 | 11 | 11 |
| 0-1 | 0 | 6 | 8 | 10 | 9 | 8 | 9 | 9 | 8 |
| (-1)-0 | 0 | 1 | 3 | 5 | 6 | 5 | 6 | 6 | 6 |
| (-2)-(-1) | 0 | 0 | 1 | 2 | 3 | 3 | 4 | 4 | 4 |
| (-3)-(-2) | 0 | 0 | 0 | 1 | 1 | 1 | 2 | 2 | 2 |
| < (-3) | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 2 | 2 |

Table: Probability distribution of headline inflation forecast

| Percent | 2562 | | 2563 | | | | 2564 | | |
|---|---|---|---|---|---|---|---|---|---|
| | Q2 | Q3 | Q4 | Q1 | Q2 | Q3 | Q4 | Q1 | Q2 |
| > 4.0 | 0 | 0 | 0 | 1 | 2 | 4 | 5 | 5 | 5 |
| 3.5-4.0 | 0 | 0 | 0 | 2 | 2 | 3 | 3 | 3 | 3 |
| 3.0-3.5 | 0 | 0 | 1 | 4 | 3 | 5 | 5 | 5 | 5 |
| 2.5-3.0 | 0 | 0 | 3 | 6 | 5 | 6 | 6 | 6 | 6 |
| 2.0-2.5 | 0 | 2 | 7 | 9 | 7 | 8 | 8 | 8 | 7 |
| 1.5-2.0 | 0 | 7 | 12 | 11 | 8 | 9 | 9 | 9 | 9 |
| 1.0-1.5 | 100 | 17 | 16 | 13 | 10 | 10 | 10 | 9 | 9 |
| 0.5-1.0 | 0 | 25 | 17 | 13 | 10 | 10 | 10 | 10 | 9 |
| 0.0-0.5 | 0 | 23 | 15 | 11 | 10 | 9 | 9 | 9 | 9 |
| (-0.5)-0.0 | 0 | 16 | 12 | 10 | 9 | 8 | 8 | 8 | 8 |
| (-1.0)-(0.5) | 0 | 8 | 8 | 7 | 8 | 7 | 7 | 7 | 7 |
| (-1.5)-(1.0) | 0 | 3 | 5 | 5 | 7 | 6 | 6 | 6 | 6 |
| (-2.0)-(-1.5) | 0 | 1 | 2 | 3 | 5 | 4 | 4 | 5 | 5 |
| < -2.0 | 0 | 0 | 2 | 4 | 12 | 9 | 10 | 11 | 12 |

Table: Probability distribution of core inflation forecast

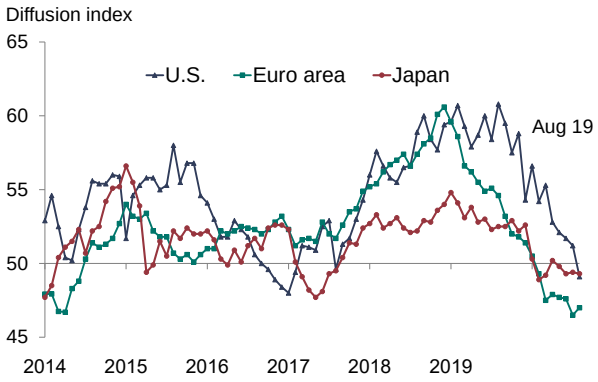
| Percent | 2562 | | 2563 | | | | 2564 | | |
|---|---|---|---|---|---|---|---|---|---|
| | Q2 | Q3 | Q4 | Q1 | Q2 | Q3 | Q4 | Q1 | Q2 |
| > 4.0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 3.5-4.0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 3.0-3.5 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2.5-3.0 | 0 | 0 | 0 | 0 | 1 | 2 | 2 | 1 | 2 |
| 2.0-2.5 | 0 | 0 | 0 | 2 | 4 | 6 | 5 | 4 | 5 |
| 1.5-2.0 | 0 | 0 | 2 | 9 | 11 | 13 | 13 | 11 | 11 |
| 1.0-1.5 | 0 | 3 | 14 | 23 | 21 | 22 | 20 | 18 | 17 |
| 0.5-1.0 | 100 | 37 | 38 | 31 | 25 | 23 | 22 | 22 | 21 |
| 0.0-0.5 | 0 | 50 | 33 | 23 | 20 | 18 | 18 | 19 | 19 |
| (-0.5)-0.0 | 0 | 10 | 12 | 10 | 11 | 10 | 11 | 13 | 13 |
| (-1.0)-(0.5) | 0 | 0 | 2 | 3 | 5 | 4 | 5 | 7 | 7 |
| (-1.5)-(1.0) | 0 | 0 | 0 | 0 | 1 | 1 | 2 | 3 | 3 |
| (-2.0)-(-1.5) | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| < -2.0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

## 4.2 Chart Pack

# Global economy

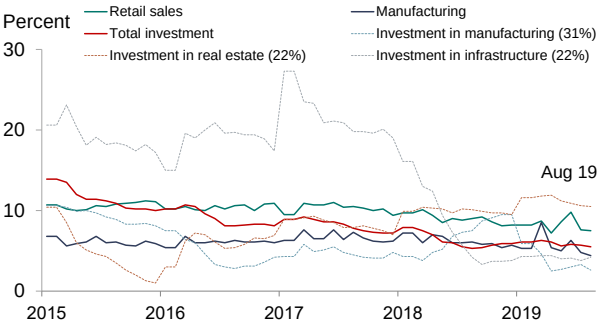
Major advanced economies were expected to moderate in line with production, exports, and private investment. Private consumption continued to be an important driver, underpinned by robust employment in the services sector. Asian economies would likely expand at a slower pace mainly on impact of trade protectionist measures.

### Manufacturing Purchasing Manager Index



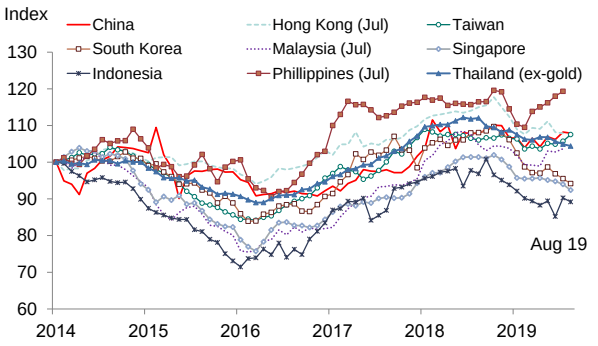

Sources: Bloomberg and Eurostat

### China's economic indicators (Change from the same period last year)



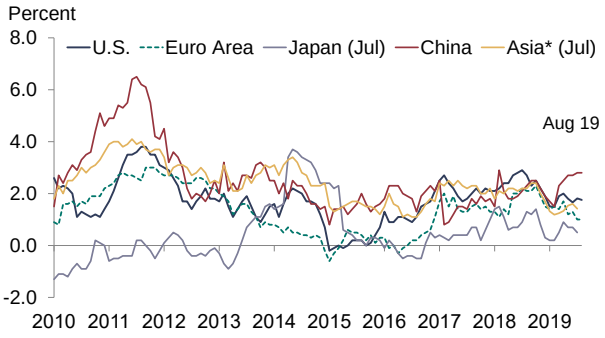

Note: ( ) denotes share to total investment
Source: CEIC

### Export value, 3-month moving average, seasonally adjusted (January 2014 = 100)




Source: CEIC

### Inflation of Thailand's major trading partners

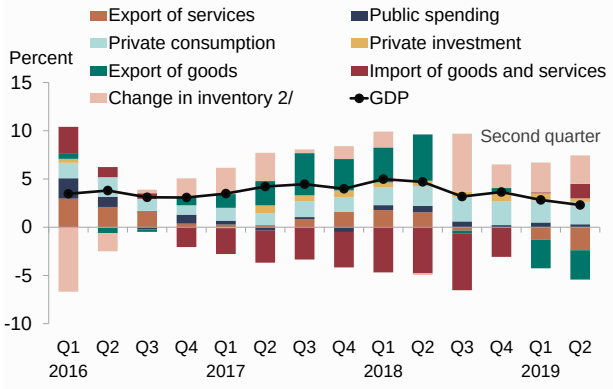



Note: *Average of headline inflation in Indonesia, South Korea, Malaysia, the Philippines, Singapore and Taiwan
Source: CEIC

# Thai economy

The Thai economy was expected to expand at a slower rate due to a slowdown in both external and domestic demand. Merchandise exports would be affected by a slowdown in trading partners' growth and global trade volume, as well as a prolonged downturn in the electronics cycle. In addition, impact of merchandise export contraction on domestic demand became more pronounced on both private consumption and investment. Meanwhile, public expenditure would continue to support the economy.

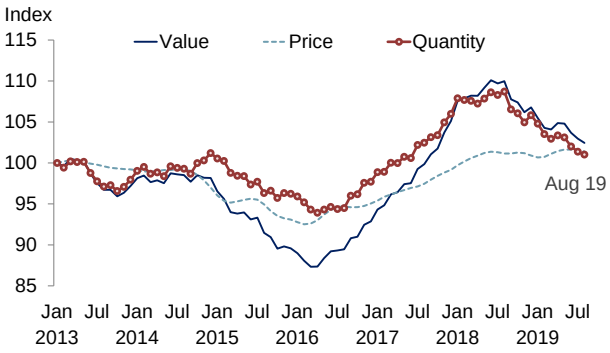
## Contribution to Thailand's GDP growth[1/]




Note: [1/] Calculated by Chain Volume Measure method (CVM)
[2/] Change in inventory and statistical discrepancy
Source: Office of National Economic and Social Development Council, calculations by the Bank of Thailand

## Thai exports (excluding gold): value, price and quantity

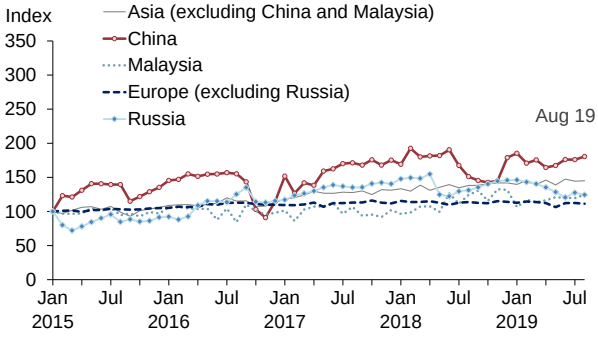
(3-month moving average, seasonally adjusted; January 2013 = 100)




Source: Customs Department and Ministry of Commerce, calculations by Bank of Thailand

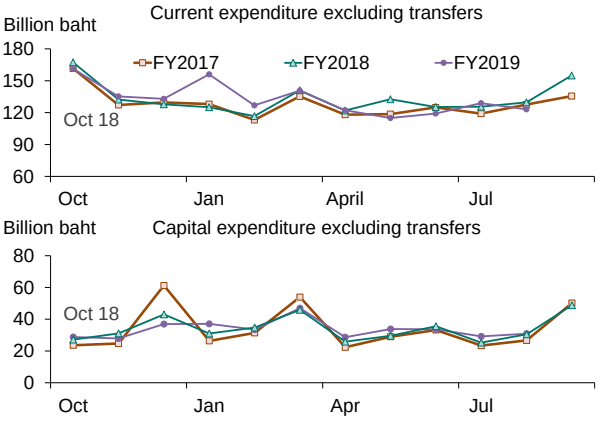
## Index of foreign tourists classified by nationality

(3-month moving average, seasonally adjusted; January 2015 = 100)




Source: Department of Tourism

## Public spending by central government

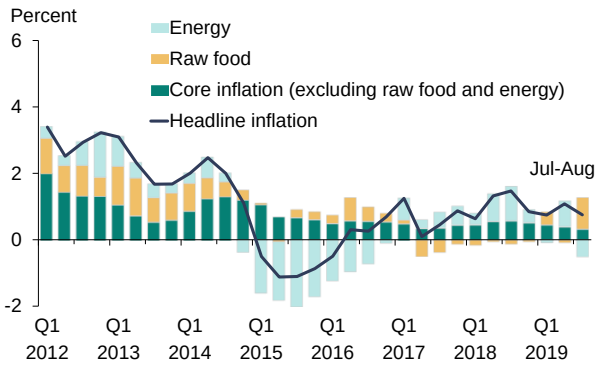



Source: Bureau of Budget, Fiscal Policy Office

# Inflation

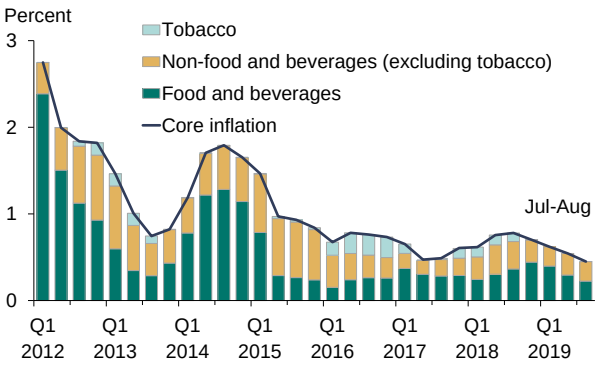
Headline inflation declined due to lower energy prices in line with global crude oil prices. Core inflation decreased mainly on account of core inflation in the food category, while prices of other goods slowly increased as a result of domestic economic slowdown and impact of structural changes. Short-term inflation expectations declined, while long-term inflation expectations remained unchanged.

### Contribution to headline inflation



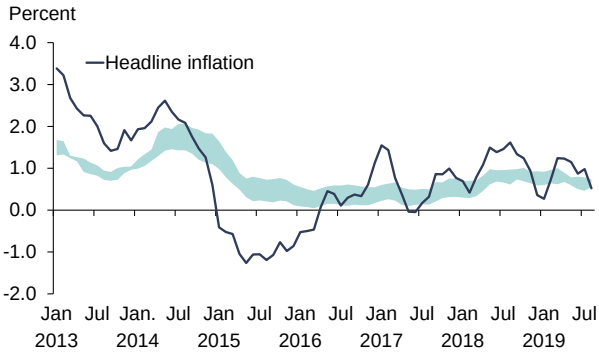

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

### Contribution to core inflation




Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

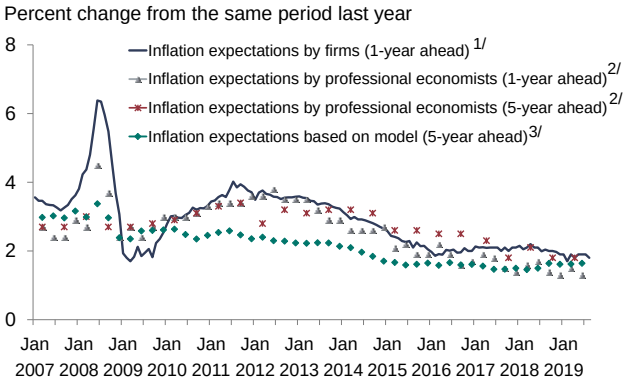
### Underlying inflation indicators




Note: The field shows the highest and lowest outcomes among different measures of underlying inflation. The measures included are 1. **Trimmed mean** (excludes goods and services with most volatile price changes, removing the bottom 15 percentile and the top 10 percentile), 2. **Principal component model** (calculates changes in common statistical components that attribute price movements across categories of goods and services) and 3. **Core inflation excluding rents and government measures.**

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

### Inflation expectations

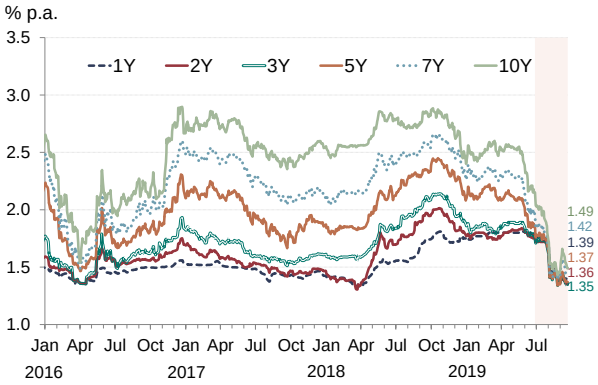



Source: 1/ Business Sentiment Survey (BSI) by the Bank of Thailand
2/ Asia Pacific Consensus Forecast
3/ Calculations based on macro-finance term structure model using bond yield and macroeconomic data

# Financial conditions

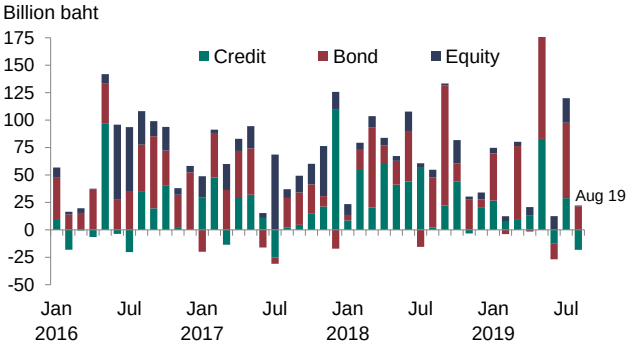
Short-term money market rates, short-term government bond yields, and commercial banks' lending rates decreased in line with the policy rate. Long-term government bond yields declined at a fast pace driven by both domestic and external factors. Financing activities in the private sector continued, particularly in the bond market. Meanwhile, financing through credit channel slowed down in tandem with moderating consumption and investment. The baht against the U.S. dollar continued to appreciate due to concerns on the global economic slowdown as well as intensifying and prolonged trade tensions. Nonetheless, the baht weakened in some periods as a result of the Bank of Thailand's measures to enhance surveillance of short-term capital inflows and the policy rate cut on 7 August 2019.

### Government bond yields




Source: Thai Bond Market Association (Thai BMA) (data as of 24 September 2019)

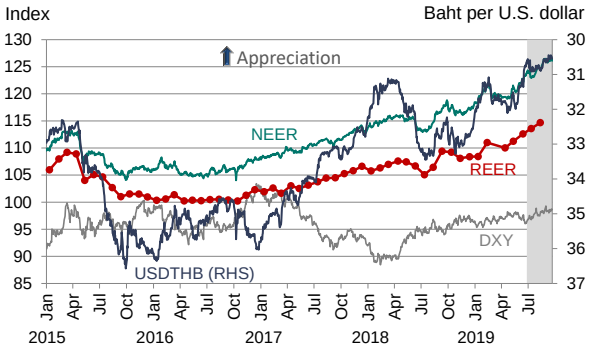
### Total corporate financing by instrument*




Note: *Monthly change in outstanding of corporate loans (seasonally adjusted), corporate bonds excluding those issued by commercial banks, and newly issued equities.

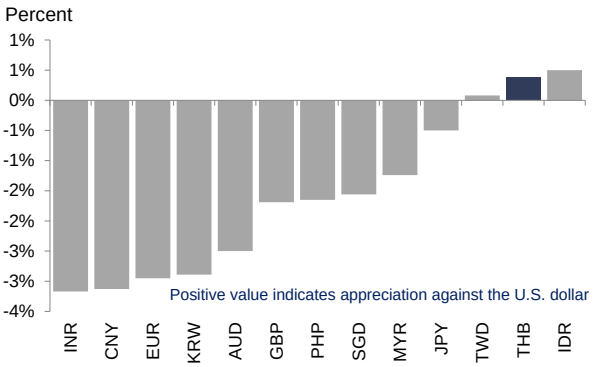
Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA)

### The Thai baht vis-a-vis U.S. dollar (USDTHB), Nominal Effective Exchange Rate (NEER), and the Dollar Index (DXY)




Source: Bank of Thailand and Reuters (data as of 24 September 2019)

### Currency movements vis-a-vis the U.S. dollar (24 September 19 compared to end of Jun 19)

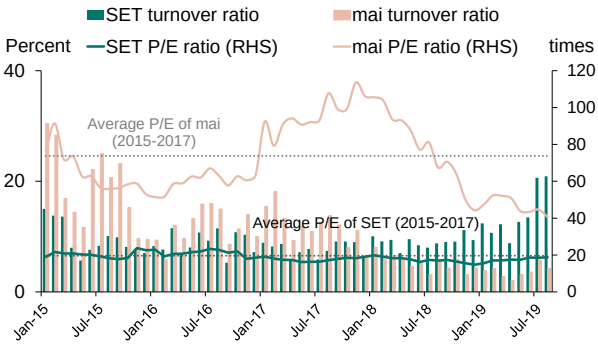



Source: Bank of Thailand and Reuters (data as of 24 September 2019)

# Stability: financial markets

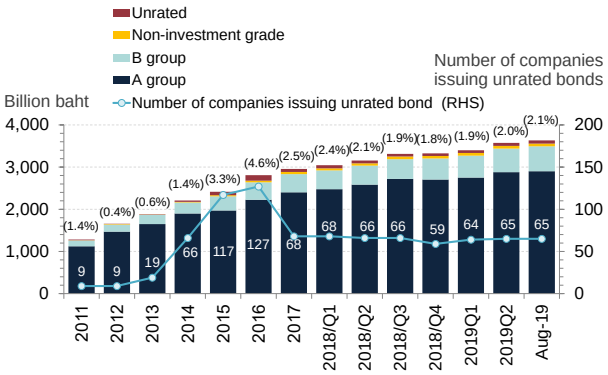
The price-to-earning (P/E) ratio of the Stock Exchange of Thailand (SET) stabilized close to its historical average, while that of the Market for Alternative Investment (mai) decreased. The share of unrated bond issuance remained stable.

Current price-to-earning ratio and turnover ratio of SET and mai




Source: Stock Exchange of Thailand (as of September 2019)

Corporate bonds outstanding

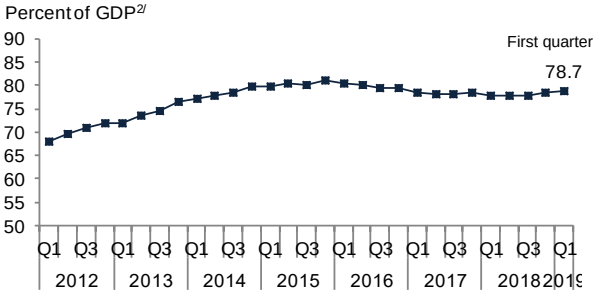



Note: ( ) represents percentage of unrated bonds to total corporate bonds
Source: Thai Bond Market Association (Thai BMA)

# Stability: household sector

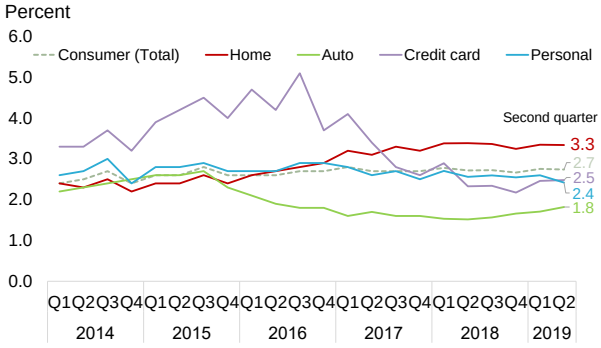
The ratio of household debt to GDP in the first quarter of 2019 rose from the previous quarter, and this trend would continue. Overall credit quality of households, as indicated by the NPL ratio, remained stable, despite signs of an increase in special mention loans (SM) across all loan types.

Household debt[1/]




Note: [1/] Loans to households by financial institutions
[2/] Calculated by averaging the 4 latest quarterly GDP
[3/] Household debt and GDP data are revised. This results in the different debt to GDP ratios compared to the last MPR.
Source: Bank of Thailand

Share of non-performing loans (NPL) in consumer loans, classified by loan type

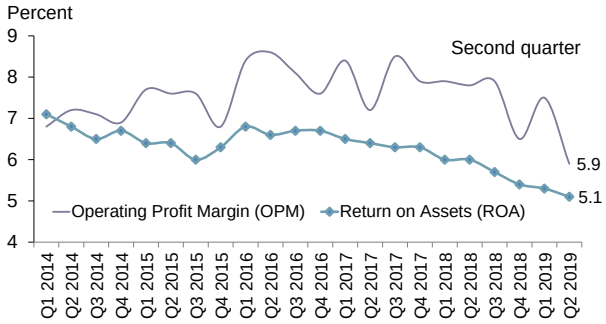



Source: Bank of Thailand

# Stability: corporate sector

Contraction in merchandise exports and baht appreciation during the second quarter of 2019 began to impact the Thai corporate sector, as reflected by deteriorating profitability and debt serviceability, especially those with a large proportion of revenue from exports. Moreover, there remained a need to monitor small businesses in certain sectors, such as construction and commercial sectors, which faced structural issues and subsequent weakness in financial positions.

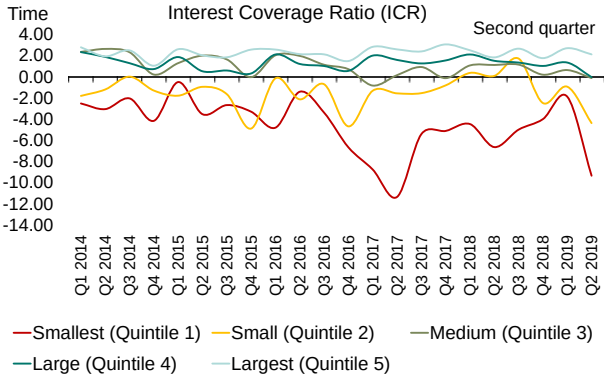
## Operating Profit Margin (OPM) and Return on Assets (ROA)*




Note: * Median estimates; ROA is returns to average assets.
OPM is operating profits to total sales.

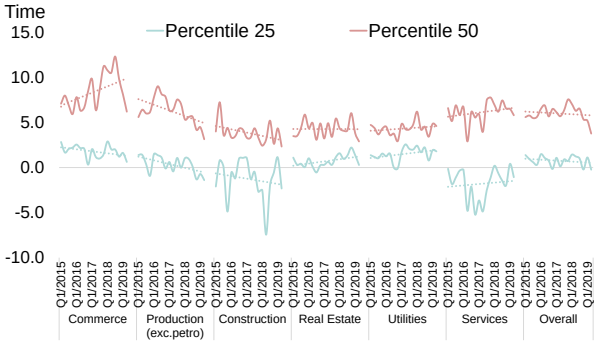
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

## Debt serviceability at 25th percentile of each group of firm size




Source: Stock Exchange of Thailand, calculation by Bank of Thailand

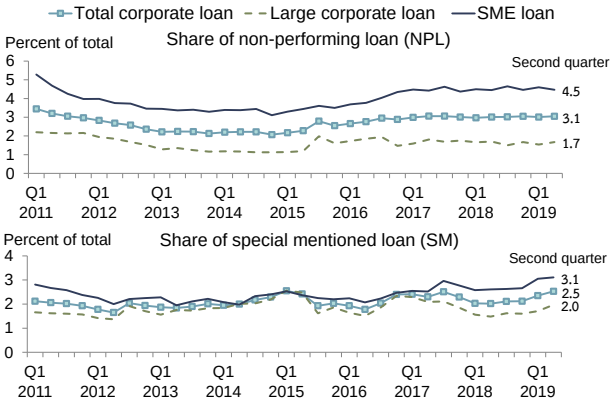
## Interest Coverage Ratio, classified by sector




Note: * production exclude Petroleum and chemicals

Source: Stock Exchange of Thailand, calculation by Bank of Thailand

## Loan quality of corporate sector

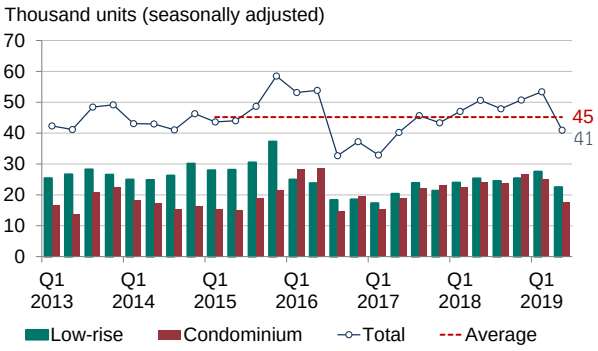



Source: Bank of Thailand

# Stability: real estate

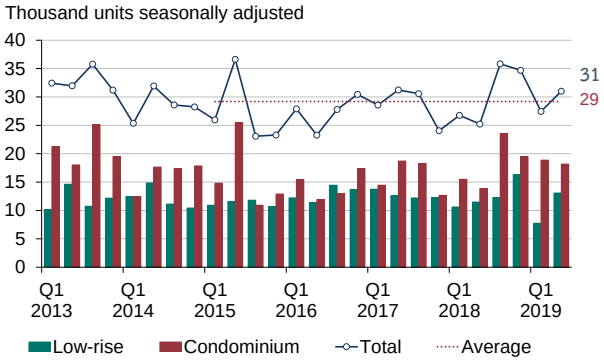
Risks of oversupply in the real estate market heightened, especially for condominiums with value exceeding 5 million baht and those along the MRT Blue Line, whose excess supply and time-to-go increased. Moreover, real estate developers in such areas continued to launch new projects. Meanwhile, ownership transfers and overall residential prices declined.

### Residential transfer units in Bangkok and vicinities



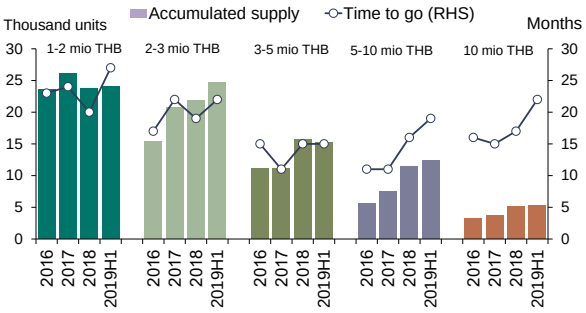

Source: Real Estate Information Center
Note: *Monthly average during 2014-2018

### New residential projects launched in Bangkok and vicinities



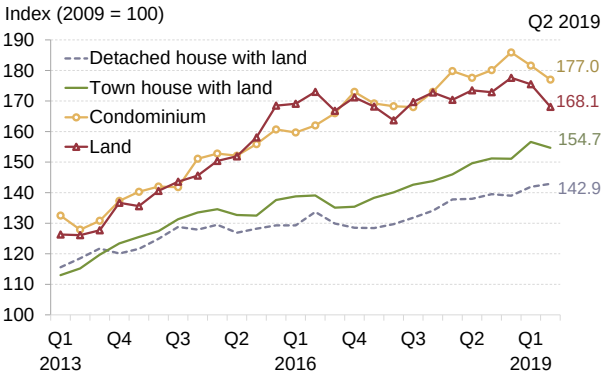

Source: Agency for Real Estate Affairs (AREA), calculation by Bank of Thailand
Note: *Monthly average during 2014-2018

### Condominium inventory in Bangkok and vicinities and 'Time to go'




Note: 'Time to go' is the time taken for all real estate inventory to be sold out at the average sales rate per month (since projects launched) given no additional supply.
Source: AREA and calculation by the Bank of Thailand

### Real estate price indices

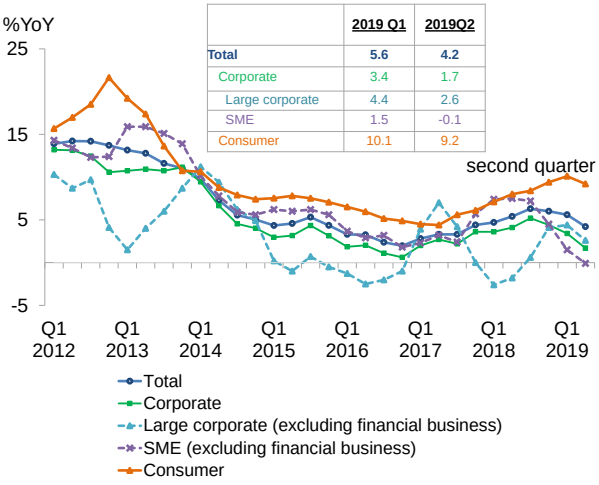



Note: Calculation based on commercial bank loan data
Source: Bank of Thailand

# Stability: financial institutions

Financial institutions maintained strong financial positions, as reflected in high levels of capital buffers among commercial banks to cushion against risks should credit quality deteriorate. In the second quarter of 2019, commercial bank credits expanded at a slower rate in line with softening economic growth outlook. Meanwhile, overall NPL ratio remained at a high level.

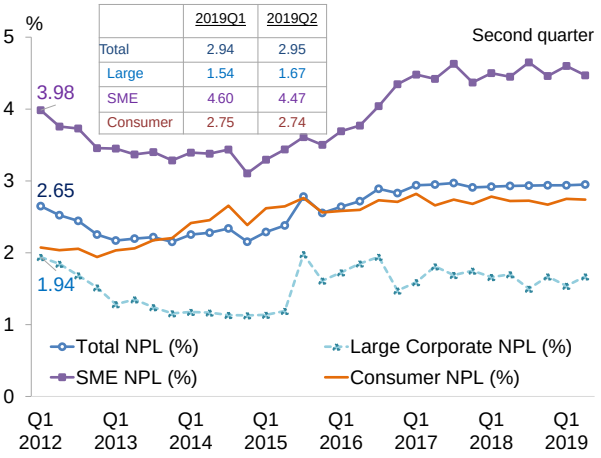
## Credit growth in the commercial bank system




| | 2019 Q1 | 2019Q2 |
|---|---|---|
| Total | 5.6 | 4.2 |
| Corporate | 3.4 | 1.7 |
| Large corporate | 4.4 | 2.6 |
| SME | 1.5 | -0.1 |
| Consumer | 10.1 | 9.2 |

Source: Bank of Thailand

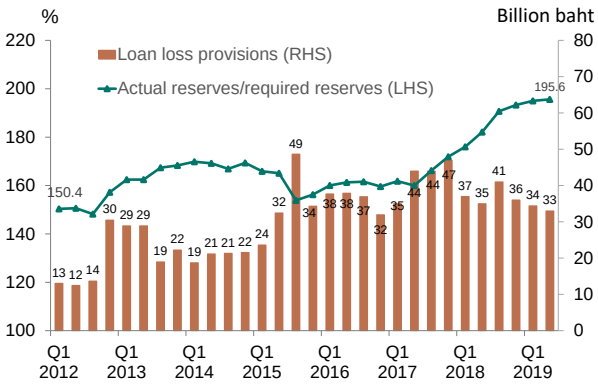
## Non-performing loan (NPL)




| | 2019Q1 | 2019Q2 |
|---|---|---|
| Total | 2.94 | 2.95 |
| Large | 1.54 | 1.67 |
| SME | 4.60 | 4.47 |
| Consumer | 2.75 | 2.74 |

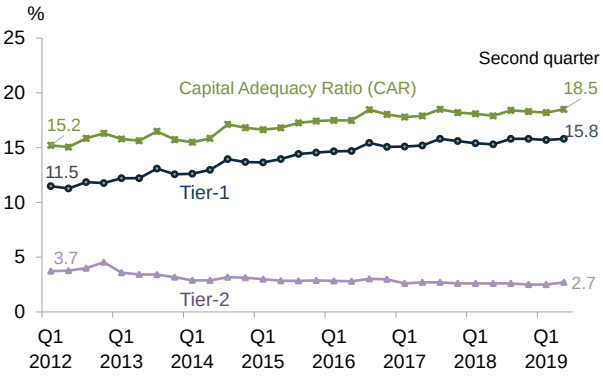
Source: Bank of Thailand

## Provisions in commercial bank system




Source: Bank of Thailand

## Capital buffers in commercial bank system

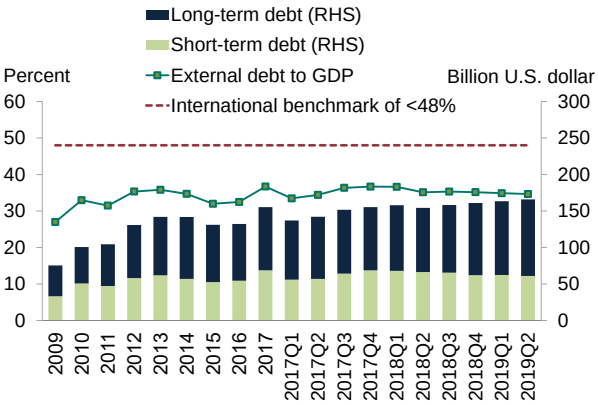



Source: Bank of Thailand

# Stability: external positions

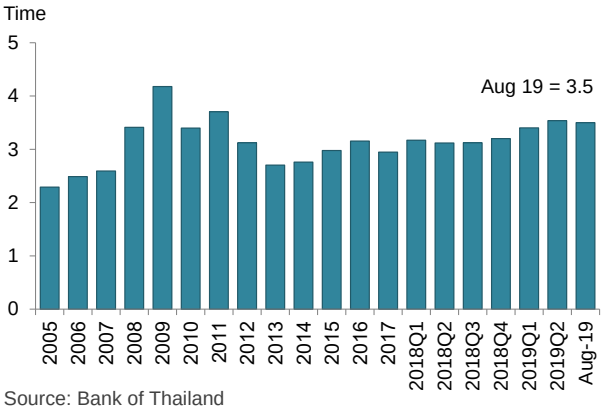
Thailand's external stability remained sound, as reflected in high levels of international reserves and sustained current account surplus. Moreover, the ratio of external debt to GDP was below an international benchmark. This would help the Thai economy to be resilient against volatilities in the global financial market.

### Thailand's external debt




Source: Bank of Thailand

### Reserve to short-term debt

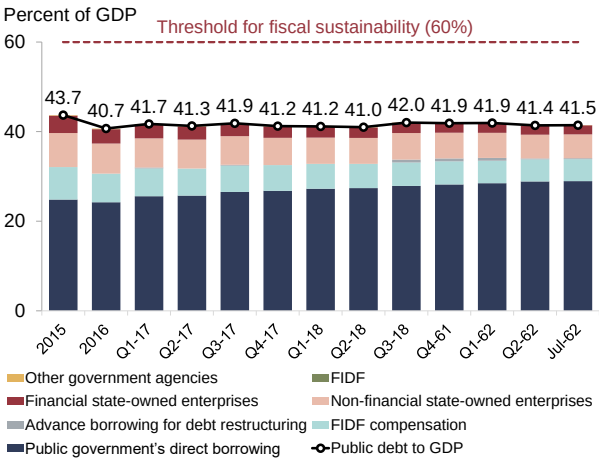



Source: Bank of Thailand

# Stability: fiscal sector

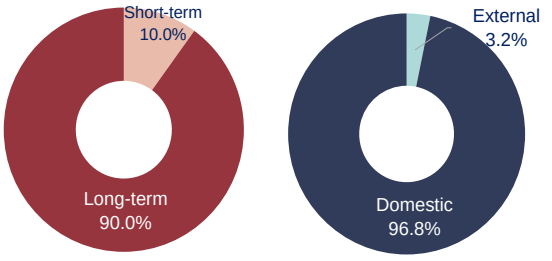
Fiscal stability remained sound. The ratio of public debt to GDP was still below the fiscal sustainability threshold.

### Public debt to GDP




Note: Calculated by GDP with Chain Volume Measure
Source: Public Debt Management Office

### Outstanding debt as of July 2019




Note: Share of short-term and long-term debt calculated from remaining duration until maturity
Source: Public Debt Management Office


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